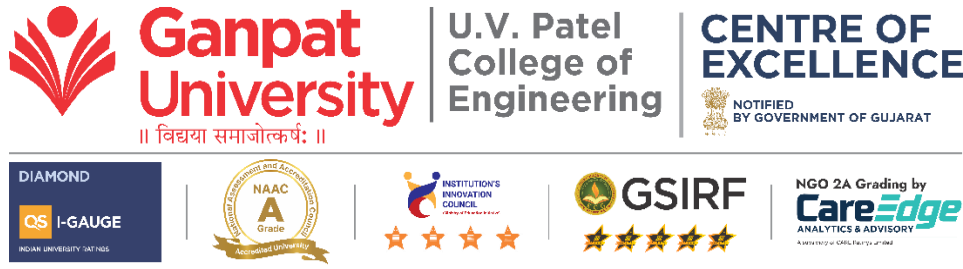


Minor Project Report on

INTELLIQUIZ - AI POWERED QUIZ SYSTEM

at

U. V. Patel College of Engineering



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U.V. PATEL COLLEGE OF ENGINEERING



22/11/2025

CERTIFICATE

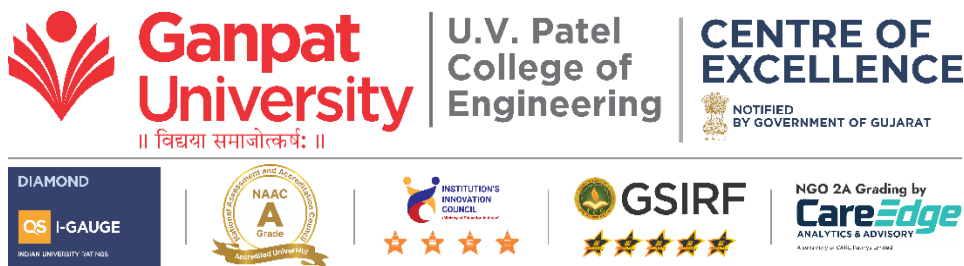
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This is to certify that Mr. Vishwarajsinh Vaghela (22012021025) student of **B.Tech. Semester VII (Information Technology)** has completed his full semester Minor Project work titled “**IntelliQuiz – AI Powered Quiz System**” satisfactorily in partial fulfillment of the requirement of Bachelor of Technology degree of Information Technology of Ganpat University, Ganpat Vidyanagar-384012, Gujarat in the year 2025-2026.

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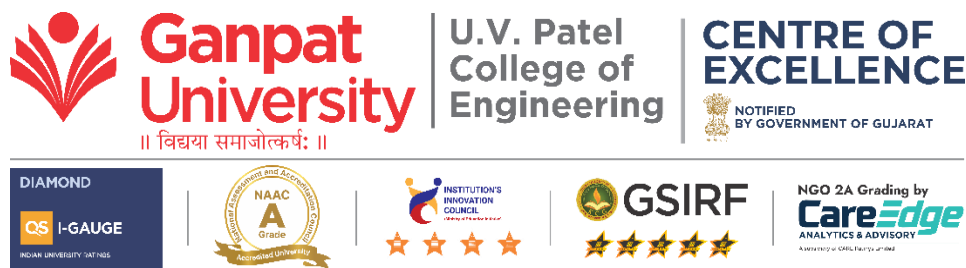
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This is to certify that Mr. Amitkumar Sathvara (22012021020) student of **B.Tech. Semester VII (Information Technology)** has completed his full semester Minor Project work titled “**IntelliQuiz – AI Powered Quiz System**” satisfactorily in partial fulfillment of the requirement of Bachelor of Technology degree of Information Technology of Ganpat University, Ganpat Vidyanagar- 384012, Gujarat in the year 2025-2026.

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This is to certify that Mr. Bhavysinh Zala (22012021027) student of **B.Tech. Semester VII (Information Technology)** has completed his full semester Minor Project work titled “**IntelliQuiz – AI Powered Quiz System**” satisfactorily in partial fulfillment of the requirement of Bachelor of Technology degree of Information Technology of Ganpat University, Ganpat Vidyanagar- 384012, Gujarat in the year 2025-2026.

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With Sincere Regards from,
Mr.Vishwarajsinh Vaghela
Mr.Amit Sathvara
Mr.Bahvyasinh Zala

ABSTRACT

The IntelliQuiz - AI Powered Quiz System is an intelligent, web-based platform designed to automate and enhance online assessments through artificial intelligence and secure role-based management. Developed using Flask in Python, the system integrates Google Gemini AI to generate topic-based multiple-choice and viva questions automatically, reducing manual effort and ensuring content diversity. It features robust anti-cheating mechanisms such as random question sequencing, plagiarism detection, time restrictions, and duplicate login prevention to maintain examination integrity. With dedicated dashboards for administrators, teachers, and students, the platform enables smooth user management, quiz creation, real-time performance tracking, and automated email notifications for credential sharing. Using SQLite as its backend and secure configuration via environment variables, IntelliQuiz ensures data protection, scalability, and reliability. By combining AI-driven automation, secure authentication, and smart analytics, the system delivers an efficient, transparent, and user-friendly solution for conducting digital assessments in modern educational environments.

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Mission	Seek, search and offer programs that lead to symbiotic emergence of 'academic excellence' and 'industrial relevance' in education and research.

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Vision	Department aims to achieve its recognition as a leading contributor in the area of technical education of computer engineering by practicing latest principles, tools and technologies to cope with current and future challenges and hence contributing to global welfare.
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1. INTRODUCTION

The IntelliQuiz - AI Powered Quiz System is an intelligent web-based platform designed to automate the creation, management, and evaluation of online quizzes using artificial intelligence. Developed with Python Flask and integrated with Google Gemini AI, it can automatically generate topic-based multiple-choice and viva questions, reducing teachers' workload and ensuring variety and accuracy in assessments. The system provides secure, role-based access for administrators, teachers, and students, each with dedicated features for managing users, creating quizzes, and viewing results. To maintain fairness, IntelliQuiz includes built-in anti-cheating measures such as random question sequencing, plagiarism detection, time restrictions, and duplicate login prevention. With a lightweight SQLite database and secure environment configuration, it offers a reliable, scalable, and user-friendly solution for conducting digital assessments efficiently in modern educational environments.

1.1 Project Overview

The IntelliQuiz - AI Powered Quiz System is a smart and secure online quiz management platform designed to simplify and automate the process of digital assessments for educational institutions. It enables teachers to create and manage quizzes effortlessly by utilizing Google Gemini AI to automatically generate multiple-choice and viva questions based on specific topics, ensuring diverse and high-quality content. Built using the Flask framework in Python, the system provides dedicated dashboards for administrators, teachers, and students, each tailored to their specific roles and responsibilities. Administrators can manage users, import data, and send credentials through automated emails, while teachers can design quizzes, monitor student performance, and track activity logs. Students can conveniently participate in quizzes, view results, and assess their progress. To maintain exam integrity, the platform integrates various anti-cheating features such as random question sequencing, plagiarism detection, time limits, and duplicate login prevention. With data securely stored in an SQLite database and sensitive configurations managed through environment variables, IntelliQuiz ensures efficiency, scalability, and security. Overall, the system provides a comprehensive, AI-driven solution that enhances the accuracy, fairness, and effectiveness of online learning and assessments.

1.2 Purpose

The primary purpose of the IntelliQuiz - AI Powered Quiz System is to provide an

intelligent, automated, and secure platform for conducting online assessments efficiently and fairly. It aims to eliminate the limitations of traditional manual examination methods by integrating artificial intelligence to generate high-quality, topic-based quiz questions automatically, thereby saving educators valuable time and ensuring question diversity and accuracy. The system is designed to streamline the entire assessment process—from quiz creation and student participation to result evaluation and reporting—within a unified, user-friendly interface. Additionally, it promotes academic integrity through built-in anti-cheating mechanisms such as random question ordering, plagiarism detection, and time-bound restrictions. By offering role-based access for administrators, teachers, and students, the IntelliQuiz System ensures organized management, transparency, and security in digital education. Ultimately, its purpose is to enhance learning experiences, reduce administrative workload, and create a smarter, technology-driven solution for modern educational environments.

1.3 Objectives

The main objective of the IntelliQuiz - AI Powered Quiz System is to develop a smart, automated, and secure platform that simplifies the creation, management, and evaluation of online quizzes using artificial intelligence. The system aims to assist educators by automatically generating topic-based questions through Google Gemini AI, thereby reducing manual effort and improving the quality and diversity of assessments. It also seeks to ensure fairness and integrity in examinations by implementing advanced anti-cheating features such as randomization of questions, plagiarism detection, time limits, and duplicate login prevention. Another key objective is to provide role-based access control, enabling administrators, teachers, and students to perform their specific tasks efficiently within a single integrated system. By securely managing data through an SQLite database and automating communication via email notifications, the project strives to create a reliable, user-friendly, and scalable platform that enhances transparency, boosts productivity, and supports the advancement of digital education and assessment practices.

2. PROJECT SCOPE

This chapter outlines the scope of the IntelliQuiz - AI Powered Quiz System project, which is designed to transform the way educational institutions conduct and manage online assessments. With the growing demand for automation, accuracy, and security in modern education, the system leverages artificial intelligence and web-based technologies to provide an efficient, intelligent, and user-friendly quiz management solution. By integrating Google Gemini AI, the platform can automatically generate topic-based multiple-choice and viva questions, significantly reducing manual workload and ensuring high-quality, diverse assessments. The system supports role-based functionality for administrators, teachers, and students, enabling seamless management of users, quizzes, and results within a secure environment. Additionally, the platform incorporates advanced anti-cheating mechanisms such as random question sequencing, time restrictions, plagiarism detection, and duplicate login prevention to ensure fairness and integrity during examinations. Designed with scalability and flexibility in mind, the IntelliQuiz System aims to streamline the academic evaluation process, enhance transparency, and promote smarter, AI-driven digital learning experiences.

2.1 Literature Review

The literature review explores existing e-learning and quiz management platforms that utilize artificial intelligence, automation, and web technologies for digital assessment and content generation. It analyzes systems used in educational and professional training environments to understand their capabilities, integration methods, and limitations. Platforms such as Google Forms, Kahoot, Quizizz, and AI-based tools like ChatGPT and OpenAI's GPT models are evaluated to assess their effectiveness in automated quiz creation, user management, and result analytics.

No.	Name of Platform	Features	Limitations
1	Flask! [1] [2]	Flask is a lightweight and flexible Python web framework that allows developers to build web applications quickly.	Flask does not include built-in components like databases, authentication, or admin panels, so developers must configure everything manually.
2	Gemini API [3]	The Gemini API provides advanced AI capabilities including text generation, structured JSON output, and reasoning abilities.	The API requires internet access, an API key, and usage may incur costs. It can occasionally generate inaccurate content or formatting errors if prompts are weak. It cannot function offline.
3	Anti-Cheating and Monitoring [4] [5]	Anti-cheating systems help secure online exams by detecting tab switching, blocking copy-paste, enforcing full screen mode, tracking answer speed,	These techniques depend heavily on browser restrictions and can be bypassed by advanced users.

Table 1 Literature Review

As shown in Table – 1 literature review summarizes the key technologies used in developing the IntelliQuiz system. It highlights the strengths and limitations of Flask as a lightweight web framework, the Gemini API for AI-based MCQ generation, and anti-cheating techniques for secure assessments. The review provides insights into how these existing tools and methods support the proposed system. It also identifies gaps in current solutions, which IntelliQuiz aims to address through automation and enhanced security.

2.2 Boundaries and Limitations

The IntelliQuiz - AI Powered Quiz System is designed to offer an intelligent and efficient solution for online quiz creation and management; however, it operates within certain boundaries and limitations. The system relies on Google Gemini AI for automatic question generation, which requires a stable internet connection and valid API access—making it dependent on external services for its AI capabilities. As it uses SQLite as the primary database, the platform is best suited for small to medium-scale institutions, with scalability limitations for very large datasets or concurrent users. Additionally, while the system includes multiple anti-cheating mechanisms such as question randomization and time-bound restrictions, it cannot fully prevent external collaboration or unauthorized device use during assessments. The email automation feature depends on SMTP configurations, which may vary across hosting environments. Moreover, AI-generated questions may occasionally require human review to ensure contextual accuracy and academic relevance. Despite these constraints, IntelliQuiz remains a robust and practical platform for enhancing digital assessments through automation, intelligence, and secure design.

2.3 Problem Statement

Traditional quiz and examination systems in educational institutions often rely on manual processes for question creation, distribution, and evaluation, leading to inefficiencies, increased workload for teachers, and limited scalability. These conventional methods lack automation, adaptability, and mechanisms to ensure fairness and integrity during online assessments. Educators often spend significant time preparing questions and managing results, while students face inconsistent evaluation standards and limited feedback mechanisms. Existing digital quiz platforms provide basic functionality but fail to integrate artificial intelligence for automatic question generation or advanced anti-cheating mechanisms for secure assessments. Moreover, many systems lack role-based access control, centralized management, and automated communication tools, which are essential for modern e-learning environments. Therefore, there is a need for an intelligent, automated, and secure quiz management system that reduces manual effort, enhances assessment quality, prevents cheating, and streamlines the overall evaluation process—addressing both administrative and academic challenges in digital education.

2.4 Functional Requirements

The functional requirements of the IntelliQuiz - AI Powered Quiz System define the core operations and features necessary for efficient quiz management and user

interaction. The system allows administrators to manage user accounts, import bulk data via CSV files, assign roles, reset passwords, and send automated email notifications with login credentials. Teachers can create and manage quizzes, generate questions automatically using Google Gemini AI, preview questions before publishing, and monitor student performance through detailed analytics and activity logs. Students can register, log in securely, participate in assigned quizzes, view their results, and track their progress over time. The system also includes essential functionalities such as role-based authentication, session management, and real-time result evaluation. Additionally, it incorporates advanced anti-cheating features like random question sequencing, plagiarism detection, time restrictions, and duplicate login prevention to maintain exam integrity. Data is stored and managed through an SQLite database, while email notifications are handled using SMTP protocols. Together, these functional requirements ensure seamless communication, automation, and secure assessment processes across all user roles.

2.4.1 User Authentication

The User Authentication module of the IntelliQuiz - AI Powered Quiz System ensures secure access control and identity verification for all users, including administrators, teachers, and students. It enables users to log in using their registered credentials, which are validated against stored records in the SQLite database, ensuring that only authorized individuals can access the system. The authentication process incorporates role-based access management, granting specific permissions based on the user's role to prevent unauthorized access to restricted features or data. The administrator can log in using predefined credentials from the .env file, while teachers and students authenticate through their registered email and password combinations. To enhance security and user experience, the system supports session management, ensuring users remain logged in securely during their active sessions while preventing multiple logins from different devices. Additionally, the system records login activities for audit and monitoring purposes. This authentication framework not only strengthens the system's security but also maintains accountability and privacy across all levels of access.

2.4.2 Quiz Making and Marking System

The Quiz Making and Marking System in the IntelliQuiz - AI Powered Quiz System Platform is designed to automate and streamline the entire assessment process—from quiz creation to result evaluation—through the integration of artificial intelligence and efficient data management. Teachers can easily create quizzes by specifying a topic, number of questions, and type (multiple-choice or viva), after which Google Gemini AI

automatically generates accurate, topic-relevant questions and answers. The system allows teachers to preview, edit, and finalize quizzes before publishing them to students. Each quiz includes customizable parameters such as time limits, question randomization, and anti-cheating configurations to ensure fairness and academic integrity. During an active quiz session, students can answer questions in real-time through an interactive and user-friendly interface. Once the quiz is submitted, the system automatically evaluates the responses using predefined correct answers stored in the SQLite database, instantly calculating scores and storing them for record-keeping. Teachers can then view detailed performance analytics, including student scores, accuracy rates, and participation logs, while students can review their results immediately. This intelligent quiz making and marking system minimizes manual effort, enhances accuracy, ensures transparency, and provides an effective and scalable.

2.4.3 Cheating Detection System

The Cheating Detection System in the IntelliQuiz - AI Powered Quiz System Platform is designed to uphold academic integrity by monitoring, preventing, and recording suspicious activities during online assessments. It incorporates multiple anti-cheating mechanisms, such as randomization of questions and options to ensure that no two students receive identical quiz patterns, and time-bound submissions to discourage prolonged or unauthorized access. The system detects duplicate logins, preventing users from accessing the same account on multiple devices simultaneously. It also monitors tab-switching behavior, which helps identify attempts to browse external sources during a quiz, and automatically logs such actions in the activity log database. Additionally, features like auto-submit on timeout, copy-paste prevention, and full screen enforcement further minimize opportunities for cheating. Teachers can view detailed logs of suspicious activities, enabling them to take necessary actions and maintain the credibility of assessments. Through these integrated detection methods, the IntelliQuiz System ensures fairness, reliability, and transparency in online examinations.

2.4.4 User Management

The User Management module in the IntelliQuiz - AI Powered Quiz System enables administrators to efficiently handle all user-related operations, ensuring organized access and smooth interaction among administrators, teachers, and students. This module allows the administrator to create, update, and delete user accounts, assign specific roles, and manage login credentials securely through the integrated database. It also supports bulk user import using CSV files, simplifying the process of adding multiple students or teachers at once. The system automatically generates secure

passwords for new users and sends their login credentials via automated email notifications using SMTP integration. Teachers can view and manage their students' profiles and performance, while students can update limited personal details related to their accounts. The user management feature also maintains role-based access control, ensuring that each user can only access functions relevant to their role. By providing centralized control, streamlined onboarding, and secure communication, the User Management module ensures efficient administration and smooth coordination within the IntelliQuiz platform.

2.5 Non-Functional Requirements

The Non-Functional Requirements of the IntelliQuiz - AI Powered Quiz System define the quality attributes and performance standards that ensure the system operates efficiently, securely, and reliably across different environments. The system is designed for high usability, featuring a simple, intuitive, and responsive user interface accessible on desktops, laptops, and mobile devices. It ensures data security through encrypted credential storage, environment variable protection in the .env file, and secure session management to prevent unauthorized access. The platform provides reliability and scalability, supporting multiple concurrent users while maintaining stable performance, particularly for small to medium-sized educational institutions. Performance efficiency is achieved through optimized database queries, fast response times, and automated processing of AI-generated questions and results. Additionally, the system emphasizes maintainability and extensibility, allowing easy updates, debugging, and integration with future technologies such as Learning Management Systems (LMS). Portability ensures smooth deployment across various operating systems and hosting environments. Collectively, these non-functional requirements guarantee that the IntelliQuiz System delivers a secure, responsive, and scalable solution for digital learning and online assessments.

2.5.1 Usability

The Usability aspect of the IntelliQuiz - AI Powered Quiz System focuses on delivering a simple, intuitive, and user-friendly experience for all types of users—administrators, teachers, and students. The system is designed with a clean and responsive web interface that ensures easy navigation, accessibility, and clarity of information. Each user role has a dedicated dashboard tailored to their specific tasks, minimizing complexity and improving workflow efficiency. Teachers can effortlessly create quizzes and analyze results, while students can easily participate in assessments and

review their performance. Clear labeling, consistent layouts, and automated feedback features enhance overall interaction and understanding. The platform's compatibility across devices such as laptops, tablets, and smartphones ensures flexibility and convenience in use. Moreover, the integration of automation, such as AI-driven question generation and auto-grading, reduces manual effort and enhances the overall user experience. By prioritizing simplicity, accessibility, and functionality, the IntelliQuiz System ensures high usability that supports effective teaching, learning, and system management.

2.5.2 Scalability

The Scalability of the IntelliQuiz - AI Powered Quiz System ensures that the platform can efficiently handle increasing workloads, users, and data volumes without compromising performance or stability. Built using the Flask framework and supported by a modular architecture, the system can be easily expanded to accommodate additional functionalities, user roles, or institutional requirements. The use of SQLite provides a lightweight database structure suitable for small to medium-scale institutions, with the flexibility to migrate to more robust databases like MySQL or PostgreSQL for larger deployments. The system's design supports multiple concurrent users and quizzes, ensuring smooth operation even during high activity periods. Its AI-based quiz generation and automated result processing are optimized to function efficiently as the number of users grows. Furthermore, the platform's architecture allows integration with cloud services or Learning Management Systems (LMS) to support institutional expansion. Overall, IntelliQuiz's scalable design ensures long-term adaptability, reliability, and consistent performance as educational needs evolve.

2.5.3 Availability

The Availability of the IntelliQuiz - AI Powered Quiz System ensures that the platform remains accessible and operational for users at all times with minimal downtime. The system is designed to provide continuous service through a reliable Flask-based web framework, ensuring stable performance during high user activity such as peak exam periods. Its lightweight SQLite database and optimized backend processes enable fast data retrieval and reduced server load, enhancing responsiveness and uptime. Proper error handling, automated database commits, and backup mechanisms further contribute to maintaining consistent availability and data integrity. Additionally, the system can be deployed on cloud servers or local networks, providing flexibility and ensuring access across different devices and geographical locations.

3. FEASIBILITY ANALYSIS

3.1 Feasibility Analysis

The Feasibility Analysis of the IntelliQuiz - AI Powered Quiz System demonstrates that the project is both practical and achievable in terms of technical, operational, and economic aspects. Technically, the system is developed using reliable and widely adopted technologies such as Python Flask, SQLite, and Google Gemini AI, ensuring compatibility, scalability, and ease of integration with existing infrastructures. The lightweight architecture enables smooth performance on standard hardware and hosting environments, requiring minimal computational resources. Operationally, the system is user-friendly, allowing administrators, teachers, and students to interact efficiently through intuitive dashboards and automated processes such as AI-driven question generation and email notifications. Economically, the solution is cost-effective due to its open-source tools and minimal server requirements, making it suitable for small to medium educational institutions with limited budgets. The integration of AI enhances productivity by reducing manual workload, while the system's modular design allows for future enhancements and scalability. Therefore, the IntelliQuiz System is highly feasible to implement, offering a sustainable, efficient, and intelligent solution for modern digital assessments.

3.1.1 Technical Feasibility

The Technical Feasibility of the IntelliQuiz - AI Powered Quiz System confirms that the project can be successfully developed and deployed using readily available and reliable technologies. The system is built on the Python Flask framework, which provides flexibility, scalability, and ease of web application development. It integrates Google Gemini AI for automated question generation, enabling intelligent and dynamic content creation based on user-defined topics. The use of SQLite as the database ensures lightweight yet efficient data storage and management, with the option to upgrade to more powerful databases like MySQL or PostgreSQL for larger-scale deployments. The frontend, designed with HTML, CSS, and JavaScript, ensures a responsive and user-friendly interface accessible across multiple devices and platforms. The integration of SMTP protocols for automated email notifications and the use of environment variables in a .env file for secure configuration management further strengthen the system's technical foundation. Overall, the chosen technologies are well-supported, cost-effective, and compatible, making the IntelliQuiz System technically feasible, robust, and adaptable for real-world educational environments.

3.1.2 Economic Feasibility

The Economic Feasibility of the IntelliQuiz - AI Powered Quiz System highlights that the project is highly cost-effective and practical for educational institutions and training organizations. The system is developed using open-source technologies such as Python Flask, SQLite, and HTML/CSS/JavaScript, which significantly reduce software licensing and development costs. The integration of Google Gemini AI provides advanced automation capabilities without the need for expensive proprietary solutions, while its pay-as-you-go API model ensures affordability based on usage. The platform's lightweight architecture minimizes hardware and hosting expenses, making it suitable for deployment on standard servers or cloud environments with minimal resource requirements. Additionally, the automated quiz generation, email communication, and result evaluation features reduce manual workload and administrative costs for teachers and institutions. Maintenance and scalability are also economically viable, as the system can be easily updated or expanded without major financial investment. Overall, the IntelliQuiz System offers a high return on investment by delivering intelligent automation, operational efficiency, and long-term cost savings, making it an economically feasible solution for modern digital education.

3.1.3 Operational Feasibility

The Operational Feasibility of the IntelliQuiz - AI Powered Quiz System ensures that the platform can be effectively implemented, managed, and utilized by administrators, teachers, and students with minimal technical expertise. The system is designed with a user-friendly interface and role-based access control, allowing each user to easily navigate and perform their respective functions, such as quiz creation, participation, and performance tracking. Teachers can generate AI-powered quizzes instantly using Google Gemini AI, while administrators can efficiently manage users, roles, and communications through automated tools. The system's automation in quiz generation, result evaluation, and email notifications significantly reduces manual workload and human error, improving institutional productivity. It can be seamlessly integrated into existing academic workflows with minimal training, ensuring quick adoption by users. Additionally, the platform's lightweight and responsive design enables smooth operation across various devices and browsers. Overall, the IntelliQuiz System is operationally feasible as it combines simplicity, automation, and reliability to enhance the effectiveness of online assessments in educational environments.

3.1.4 Implementation Feasibility

The Implementation Feasibility of the IntelliQuiz - AI Powered Quiz System

demonstrates that the project can be efficiently deployed and maintained with minimal complexity and cost. The system's modular architecture, developed using Python Flask, allows for straightforward setup, customization, and integration with existing institutional infrastructure. It can be easily hosted on local servers or cloud-based environments, requiring only basic hardware and internet connectivity. The use of SQLite ensures quick database initialization without complex configurations, while the integration of Google Gemini AI and SMTP email services can be achieved through simple API key and environment variable settings in the .env file. The platform's web-based design eliminates the need for dedicated installations, enabling users to access it directly through any browser. Furthermore, clear documentation and straightforward configuration steps make the deployment process manageable even for institutions with limited technical expertise. Maintenance and updates can be performed with ease due to the system's clean code structure and modular components. Thus, the IntelliQuiz System is highly feasible to implement, offering a practical, scalable, and low-maintenance solution for intelligent online assessments.

4. SOFTWARE REQUIREMENT SPECIFICATIONS

The Software Requirement Specifications (SRS) for the IntelliQuiz - AI Powered Quiz System define the functional and non-functional requirements essential for developing a reliable, secure, and intelligent online quiz management platform. The system is built using Python Flask as the backend framework and integrates Google Gemini AI for automated question generation based on user-defined topics. It uses SQLite as the database for storing user details, quizzes, questions, and results, ensuring efficient and lightweight data management. The platform supports three primary user roles—Administrator, Teacher, and Student—each with distinct privileges such as user management, quiz creation, and participation. It requires an active internet connection for AI integration, email notifications, and cloud-related functionalities. The frontend is developed using HTML, CSS, and JavaScript, providing a responsive and user-friendly interface. Non-functional requirements include data security through environment variables stored in a .env file, system scalability for small to medium institutions, cross-platform compatibility, and robust error handling. Overall, the SRS outlines a system designed to deliver automation, accuracy, and integrity in digital assessments while ensuring usability, performance, and security for all users.

4.1 Hardware Requirements for Developer

The Hardware Requirements for Developers of the IntelliQuiz - AI Powered Quiz System are minimal and designed to support smooth development, testing, and deployment of the application. Developers require a computer system with at least an Intel Core i5 processor or equivalent, 8 GB of RAM, and a minimum of 256 GB of storage to efficiently run development tools, virtual environments, and local servers. A 64-bit operating system such as Windows, Linux, or macOS is recommended to ensure compatibility with Python and Flask frameworks. Additionally, a stable internet connection is essential for accessing APIs like Google Gemini AI, installing dependencies, and managing version control through GitHub or similar platforms. A modern web browser (such as Google Chrome, Mozilla Firefox, or Microsoft Edge) is necessary for testing the web interface, while tools like Visual Studio Code, Postman, and SQLite Browser are beneficial for development and debugging. For email automation and AI integration testing, a functional SMTP service and valid API credentials are also required. Overall, the hardware requirements for developers are moderate and easily achievable, ensuring an efficient and cost-effective development environment for the IntelliQuiz System.

4.2 Software Requirements for Developer

The Software Requirements for Developers of the IntelliQuiz - AI Powered Quiz System include all necessary tools, frameworks, and dependencies needed to design, develop, test, and deploy the application efficiently. Developers must install Python 3.10 or later, which serves as the core programming language for the backend, along with essential frameworks such as Flask for web development and SQLite for database management. Supporting libraries like `python-dotenv`, `cryptography`, `google-auth`, `google-auth-oauthlib`, `requests`, and `Werkzeug` are required as specified in the `requirements.txt` file to enable secure authentication, environment configuration, and API integration. For code editing and debugging, an IDE such as Visual Studio Code, PyCharm, or Sublime Text is recommended. Additionally, Postman or similar tools can be used for API testing, while Git enables version control and collaborative development. A web browser (Chrome, Firefox, or Edge) is needed for interface testing, and SMTP configuration tools help verify automated email functionalities. Developers also need access to a valid Google Gemini API key for AI-driven quiz generation. Together, these software components create a complete, efficient, and reliable development environment for building and maintaining the IntelliQuiz System.

4.3 Hardware Requirements for End User

The Hardware Requirements for End Users of the IntelliQuiz - AI Powered Quiz System are minimal, ensuring accessibility and ease of use across a wide range of devices. End users, including administrators, teachers, and students, only need a basic computer or smart device with an Intel Core i3 processor or higher (or equivalent), at least 4 GB of RAM, and 2 GB of available storage space to run a modern web browser smoothly. Since the system is entirely web-based, no additional installations are required, and it can be accessed through browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari. A stable internet connection is necessary for online quizzes, real-time evaluation, and AI-based question generation. The platform is fully responsive, allowing users to access it from laptops, desktops, tablets, or smartphones without performance issues. For teachers and administrators managing multiple operations, a larger screen resolution (1366×768 or higher) is recommended for optimal viewing and navigation. Overall, the IntelliQuiz System is lightweight and optimized to perform efficiently on standard hardware, making it highly accessible to end users with minimal resource requirements.

4.4 Software Requirements for End User

The Software Requirements for End Users of the IntelliQuiz - AI Powered Quiz System are minimal, as the platform is entirely web-based and designed for easy accessibility without the need for complex installations. End users only require a modern web browser such as Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari to access the system's interface. The platform is compatible with all major operating systems, including Windows, Linux, macOS, Android, and iOS, ensuring seamless usability across devices. A stable internet connection is essential for accessing AI-generated quizzes, submitting responses, and viewing results in real time. For optimal performance, users should have JavaScript enabled in their browsers and up-to-date browser versions to ensure compatibility with the system's responsive design and dynamic features. No additional plugins or software installations are required, as all processing and storage occur on the server side. Overall, the IntelliQuiz System provides a simple, cross-platform, and user-friendly experience that ensures smooth operation and accessibility for all users.

5. PROCESS MODEL

The Process Model describes the overall workflow and structured approach followed during the development of the IntelliQuiz – AI Powered Quiz System. It outlines how system activities are planned, executed, and managed to ensure a smooth and organized development lifecycle. This section includes the selection and explanation of the software development methodology used in the project—such as the Waterfall Model, Iterative Model, or Agile approach—based on project requirements and constraints.

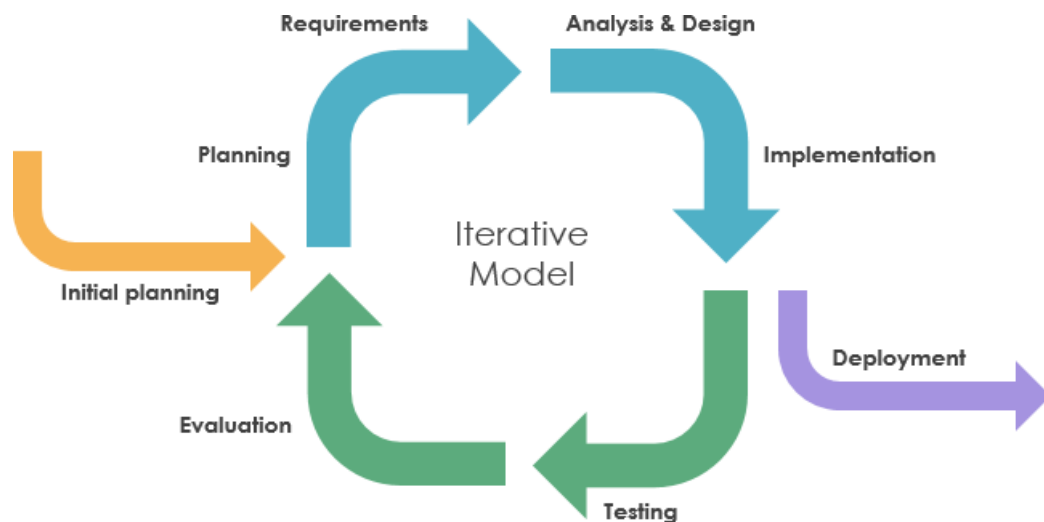


Figure 1 Iterative Model

As shown in Figure - 1 Process Model adopted for the IntelliQuiz - AI Powered Quiz System follows the Waterfall Model, ensuring a structured and systematic approach to software development. This model divides the project into sequential phases—Requirement Analysis, System Design, Implementation, Testing, Deployment, and Maintenance—where each phase must be completed before the next one begins. During the Requirement Analysis phase, both functional and non-functional needs of the system were identified, such as AI-driven quiz generation, user management, and security measures. In the System Design phase, the overall architecture, database schema, and user interface layouts were created to establish a clear development blueprint. The Implementation phase involved coding the application using Python Flask, integrating Google Gemini AI, and configuring the SQLite database and SMTP services. The Testing phase ensured that all modules, including authentication, quiz generation, and result evaluation, functioned correctly and securely. After successful validation, the system proceeded to Deployment, where it was made accessible to end users through a web server. Finally, the Maintenance phase focuses on updating AI

features, improving performance, and fixing potential bugs based on user feedback. This structured Waterfall approach ensures clarity, reliability, and efficient development of the IntelliQuiz System from conception to delivery.

5.1 Project Phases

IntelliQuiz – AI Powered Quiz System is a systematic and efficient development process, following a sequential workflow from planning to deployment. The project begins with the Requirement Analysis phase, where the system’s objectives, user roles, and AI integration needs are identified. This is followed by the System Design phase, which defines the architecture, database structure, and user interface layout to ensure functionality and scalability. The Implementation phase involves developing the application using Python Flask, integrating Google Gemini AI for automatic quiz generation, configuring the SQLite database, and setting up email automation through SMTP. Next, in the Testing phase, all system modules—including authentication, quiz creation, result evaluation, and anti-cheating mechanisms—are rigorously tested to verify accuracy, performance, and security. After successful validation, the Deployment phase makes the system accessible to end users through a web server, enabling real-world use by administrators, teachers, and students. Finally, the Maintenance and Enhancement phase ensures long-term reliability by addressing bugs, optimizing performance, and incorporating user feedback or new AI features. Together, these phases create a well-structured development lifecycle that ensures the IntelliQuiz System meets its goals efficiently and effectively.

5.1.1 Initial Planning

The Initial Planning phase of the IntelliQuiz - AI Powered Quiz System focuses on defining the project’s goals, scope, and feasibility to establish a clear direction for development. During this phase, the primary objective—to create an intelligent, automated, and secure online quiz management platform—was identified, along with the need for AI integration to simplify quiz creation and evaluation. The project team analyzed the limitations of existing systems, formulated key features such as role-based access, anti-cheating mechanisms, and real-time result generation, and determined the technologies required, including Python Flask, Google Gemini AI, and SQLite. Responsibilities were assigned among developers, designers, and testers to ensure organized workflow and effective collaboration. Additionally, a development timeline and resource plan were prepared to align with technical, financial, and operational feasibility. Risk assessment and mitigation strategies were also considered to minimize

potential challenges during implementation. Overall, this phase provided a structured foundation for the project, ensuring that all objectives, tools, and processes were clearly defined before moving into design and development.

5.1.2 Planning

The Planning phase of the IntelliQuiz - AI Powered Quiz System involved developing a detailed roadmap to guide the entire project lifecycle, ensuring that all activities were strategically organized and aligned with the system's objectives. This phase focused on outlining the technical specifications, selecting appropriate tools and technologies such as Python Flask, Google Gemini AI, SQLite, and HTML/CSS/JavaScript, and defining the tasks and timelines for each stage of development. The team established clear milestones for requirement gathering, system design, implementation, testing, and deployment while assigning specific roles and responsibilities to developers, designers, and testers. Resource allocation, including hardware, software, and human expertise, was carefully planned to ensure efficiency and minimize delays. Risk factors such as integration issues, data security, and API dependency were identified, and contingency plans were formulated to address them. Additionally, the planning phase included designing communication channels and documentation methods to maintain coordination among team members. Overall, this phase laid the groundwork for a structured, organized, and goal-oriented development process, ensuring the successful and timely execution of the IntelliQuiz System.

5.1.3 Requirements

The Requirements phase of the IntelliQuiz - AI Powered Quiz System focused on identifying, analyzing, and documenting all functional and non-functional needs essential for the system's successful operation. During this phase, the core requirements were established, including AI-driven quiz generation using Google Gemini AI, role-based user authentication and management, automated result evaluation, and email notifications for communication. Functional requirements defined what the system must do, such as quiz creation, participation, performance tracking, and anti-cheating mechanisms, while non-functional requirements outlined performance, security, scalability, and usability standards. The system needed to support multiple user roles—Administrator, Teacher, and Student—each with distinct access privileges and functionalities. Integration with AI and SMTP services was specified to ensure automation and reliability. Hardware and software requirements for both developers and end users were also determined to guarantee compatibility and smooth operation. Additionally, data security, user privacy, and efficient database management using

SQLite were prioritized to maintain system integrity. Overall, the requirements phase provided a comprehensive blueprint, ensuring that all user expectations and technical needs were clearly defined before proceeding to the design stage.

5.1.4 Analysis & Design

The Analysis and Design phase of the IntelliQuiz - AI Powered Quiz System involved transforming the gathered requirements into a structured framework and detailed design that guided the system's development. During the analysis stage, the functional components—such as user authentication, AI-powered quiz generation, result evaluation, and anti-cheating mechanisms—were thoroughly examined to determine their interactions and dependencies. Data flow diagrams, use case models, and entity-relationship diagrams were developed to visualize system operations and database relationships. In the design phase, the system architecture was structured using the Python Flask framework, defining a modular approach with clear separation between frontend, backend, and database layers. The SQLite database schema was designed to efficiently manage users, quizzes, questions, results, and activity logs, while Google Gemini AI integration was planned to handle automated question generation. The user interface was designed to be intuitive and responsive, ensuring smooth navigation for administrators, teachers, and students. Security elements such as encrypted credentials, session management, and environment variable protection were also incorporated into the design. Overall, this phase established the logical and physical structure of the IntelliQuiz System, ensuring scalability, usability, and maintainability throughout development.

5.1.5 Implementation

The Implementation phase of the IntelliQuiz - AI Powered Quiz System focused on transforming the analyzed and designed components into a fully functional software application. Using the Python Flask framework, the development team coded the backend logic, implemented database connections with SQLite, and integrated the Google Gemini AI API to enable automatic quiz question generation based on user-defined topics. Each module—such as user authentication, quiz creation, result evaluation, and email automation—was developed and tested individually to ensure functionality and reliability. Role-based access control was established to differentiate the permissions and dashboards of administrators, teachers, and students. The frontend was built using HTML, CSS, and JavaScript, ensuring a responsive and user-friendly interface compatible with multiple devices. The integration of SMTP services allowed for automated email notifications for account creation and quiz-related communication.

Throughout the implementation process, code optimization, debugging, and documentation were performed to enhance performance, security, and maintainability. This phase successfully brought the system design to life, resulting in a stable, interactive, and intelligent web-based quiz management platform.

5.1.6 Testing

The Testing phase of the IntelliQuiz - AI Powered Quiz System was conducted to ensure the accuracy, reliability, and performance of all system components before deployment. Various testing methods, including unit testing, integration testing, system testing, and user acceptance testing (UAT), were performed to verify that each module—such as user authentication, AI-based quiz generation, result evaluation, and email automation—functioned correctly and interacted seamlessly. The system was tested for functional correctness, data validation, and security vulnerabilities, ensuring that only authorized users could access specific roles and features. Performance testing was conducted to evaluate the system's responsiveness and stability under different user loads, while compatibility testing ensured smooth operation across multiple browsers and devices. Special attention was given to testing the anti-cheating mechanisms, including tab-switch detection, randomization, and duplicate login prevention, to maintain examination integrity. Any identified bugs or inconsistencies were corrected promptly, and regression testing was performed to confirm stability after fixes. Overall, this phase ensured that the IntelliQuiz System met all specified requirements, delivering a robust, secure, and efficient platform for online assessments.

5.1.7 Evaluation

The Evaluation phase of the IntelliQuiz - AI Powered Quiz System was carried out to assess the overall performance, functionality, and user satisfaction of the developed platform after testing and deployment. This phase focused on ensuring that the system met all predefined objectives, including automated quiz generation through Google Gemini AI, secure role-based access, accurate result computation, and effective anti-cheating mechanisms. Feedback was collected from test users—administrators, teachers, and students—to evaluate the system's usability, interface design, and response time. Performance metrics such as system speed, accuracy of AI-generated questions, reliability of email notifications, and database efficiency were analyzed to verify operational excellence. The evaluation also examined scalability, ensuring that the system could handle multiple quizzes and concurrent users without performance degradation. Identified areas for improvement, such as UI enhancements or feature refinements, were documented for future updates. Overall, the evaluation confirmed

that the IntelliQuiz System successfully fulfilled its intended goals, demonstrating reliability, accuracy, and efficiency as an intelligent online quiz management platform.

5.1.8 Deployment

The Deployment phase of the IntelliQuiz - AI Powered Quiz System involved installing, configuring, and launching the fully tested application for real-world use by administrators, teachers, and students. The system was deployed on a web server to ensure online accessibility through standard web browsers without the need for additional installations. During deployment, environment variables such as API keys, email credentials, and security tokens were securely configured in the .env file to maintain data privacy and system integrity. The SQLite database was initialized with necessary tables and relationships, and sample data was loaded to verify operational readiness. The deployment process also included linking the Google Gemini AI API for automated question generation and configuring SMTP services for email notifications. Thorough verification was performed to ensure smooth connectivity, proper authentication, and efficient quiz functionality. End-user training and documentation were provided to help administrators and teachers navigate and manage the system effectively. Overall, the deployment phase marked the successful transition of the IntelliQuiz System from development to a live, accessible, and fully functional platform ready for use in real academic environments.

6. PROJECT PLAN

The Project Plan Phase outlines the structured roadmap for developing the IntelliQuiz – AI Powered Quiz System. This phase defines the project timeline, major milestones, resource allocation, and workflow sequencing required to ensure smooth and timely execution. It includes detailed planning of design activities, development tasks, integration points, and testing cycles, all mapped through tools like the Gantt Chart for visual clarity.

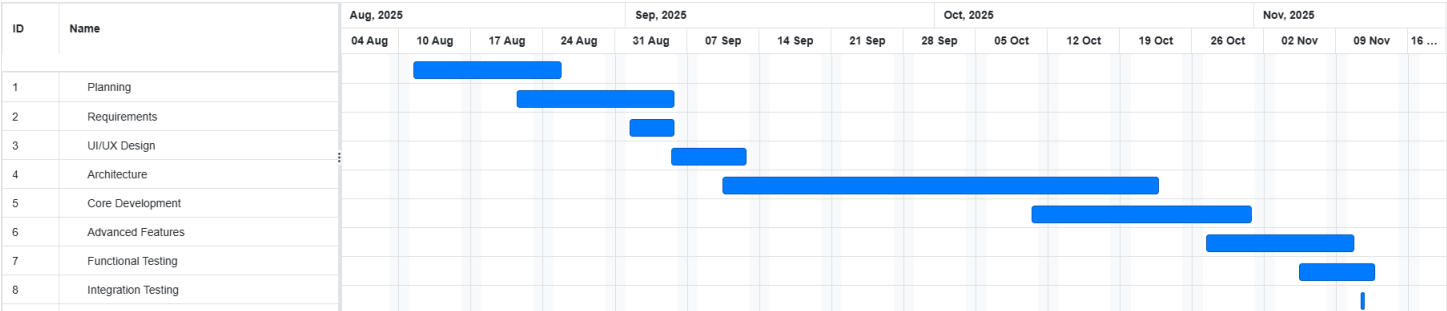


Figure 2 Gantt Chart

As shown in Figure 2, the Gantt Chart provides a clear overview of the entire project timeline, highlighting major phases from system initiation to final deployment. The chart divides the development lifecycle into three key stages: System Design, Development, and Testing. The System Design phase focuses on crafting the User Interface (UI), database schema, and the overall system architecture. This includes planning how key modules such as AI-powered question generation, quiz management, cheating detection, and user authentication will interact. Detailed workflow diagrams, API structures, and role-based system components are designed during this stage to ensure clarity and smooth integration.

6.1 System Design Phase: This stage focuses on creating the UI/UX layout, designing the database schema, and defining the overall system architecture. Key modules—including AI-based question generation, quiz management, cheating detection, and user authentication—are structured during this phase. Workflow diagrams, API frameworks, and role-based interactions are also planned to ensure smooth integration across all components. The Testing phase includes test planning, writing module-specific test cases, performing functional and non-functional testing, validating AI-generated questions, and conducting end-to-end integration tests. This phase ensures the reliability, accuracy, and security of the system across different devices, browsers, and usage scenarios.

6.2 Development Phase: The development stage is divided into core and advanced feature implementation. Core development includes functionalities such as user authentication, role-specific dashboards, quiz creation, AI-powered question generation, quiz participation, and automated scoring. Advanced development covers analytics dashboards, performance tracking, suspicious activity monitoring, and comprehensive report generation for teachers and administrators.

6.3 Testing Phase: This stage involves preparing test plans, creating detailed test cases, and conducting functional, non-functional, and integration testing. AI-generated questions are validated for accuracy, and the entire system is tested across different devices and environments to ensure security, reliability, and a consistent user experience.

Each task in the Gantt Chart is visually sequenced to reflect dependencies between activities, enabling efficient workflow management and timely progress monitoring. This structured timeline helps identify critical paths, prevent bottlenecks, optimize resource allocation, and maintain development discipline to achieve all major project milestones.

7. SYSTEM DESIGN

The System Design of the IntelliQuiz - AI Powered Quiz System defines the overall structure, architecture, and interaction of system components to ensure seamless functionality, scalability, and security. The system follows a three-tier architecture comprising the presentation layer, application layer, and database layer. The presentation layer, built with HTML, CSS, and JavaScript, provides an intuitive and responsive user interface for administrators, teachers, and students. The application layer, developed using the Python Flask framework, manages the system's core logic, including user authentication, AI-based quiz generation using Google Gemini API, quiz evaluation, and automated email communication through SMTP integration. The database layer, powered by SQLite, stores structured data such as user details, quizzes, questions, results, and activity logs, ensuring data integrity and efficient retrieval. The design also incorporates role-based access control to define user privileges and enhance system security, along with anti-cheating mechanisms such as random question sequencing and session tracking to maintain examination fairness.

7.1 Database Table

Table Name	Purpose	Key Columns
users	Store score user authentication and role information for administrators, teachers, and students.	id, username, email, password, role, created_at
students	Holds additional student-specific details and links to user accounts.	id, user_id, enrollment_number
teachers	Contains teacher-related Data,including departmental information.	id, user_id, department
admins	Stores admin-specific data and links to the main user account	id, user_id

Table Name	Purpose	Key Columns
quizzes	Manages all quiz information created by teachers, including title, room code, and time limit.	id, teacher_id, title, room_code, time_limit, anti_cheating_features, created_at
questions	Stores all quiz questions along with options and correct answers, linked to a specific quiz.	id, quiz_id, question_text, options, correct_answer
results	Records quiz results and scores of students for each quiz taken.	id, student_id, quiz_id, score, submitted_at
activity_log	Logs student activity during quizzes for anti-cheating analysis and monitoring.	id, student_id, quiz_id, action, ip, timestamp
student_answers	Stores each student's submitted answers for all quiz questions.	id, result_id, question_id, answer, timestamp

Table 2 Database Table

As shown in Table - 2, the database structure is designed to efficiently manage all quiz-related operations within the IntelliQuiz system. The quizzes table stores core quiz details such as title, room code, time limit, and associated teacher information. The questions table maintains all MCQs linked to each quiz, while the results table records student scores and submission timestamps. The activity_log table tracks student actions for anti-cheating analysis, and the student_answers table stores every selected answer for accurate evaluation.

7.2 Data Dictionary

Attribute Name	Description	Data Type	Size	Constraints
user_id	Unique identifier for each registered user in the system.	INT	10	Primary Key, Not Null

Attribute Name	Description	Data Type	Size	Constraint
username	Username chosen by the user for login purposes.	VARCHAR	50	Unique, Not Null
Email	Registered email address used for communication and login.	VARCHAR	100	Unique, Not Null
Password	Encrypted password the student	VARCHAR	255	Not Null
role	Defines the role of the user (admin, teacher, student).	VARCHAR	20	Not Null, Check(role IN ('admin', 'teacher', 'student'))
teacher_id	Unique identifier for each teacher in the system.	INT	10	Foreign Key (users.id), Not Null
student_id	Unique identifier for each student.	INTEGER	10	Foreign Key (users.id), Not Null
quiz_id	Unique identifier for each quiz created by a teacher.	INTEGER	10	Primary Key, Not Null, Auto Increment
title	Title or subject of the quiz.	VARCHAR	150	Not Null

Attribute Name	Description	Data Type	Size	Constraints
time_limit	Time duration (in minutes) allotted for completing a quiz.	INTEGER	3	Default 10, Not Null
question_id	Unique identifier for each question in	INTEGER	10	Primary Key, Not Null, Auto Increment
question_text	The text or statement of the question.	TEXT	—	Not Null
options	Stores possible answer choices in JSON or text format.	TEXT	—	Nullable
correct_answer	Index or value of the correct answer from the options list.	INTEGER	2	Nullable
score	The marks obtained by a student in a quiz.	INTEGER	2	Not Null
submitted_at	Timestamp recording when the quiz was submitted.	DATETIME	—	Default CURRENT_TIMESTAMP
activity_id	Unique identifier for each logged student activity.	INTEGER	10	Primary Key, Not Null, Auto Increment
timestamp	Time when the activity occurred	INTEGER	10	Not Null

Table 3 Data Dictionary
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As shown in Table - 3, the attributes define the structure and constraints of the quiz question and result-related fields used in the IntelliQuiz system. Each attribute specifies the purpose, data type, and validation rules required for consistent data storage. Attributes like *question_id* are primary keys with auto-increment for unique identification, while fields such as *question_text* ensure that every stored question has meaningful content.

7.3 Diagrams

To understand the working of the system more clearly, various diagrams are used to represent different aspects of the application. These diagrams help visualize how users interact with the system, how data flows between different components, and how the internal architecture is structured. They serve as a blueprint for both developers and stakeholders, making the system easier to understand, develop, and maintain. Each diagram provides a unique perspective that supports better planning, communication, and implementation throughout the project lifecycle.

7.3.1 Use-Case Diagram

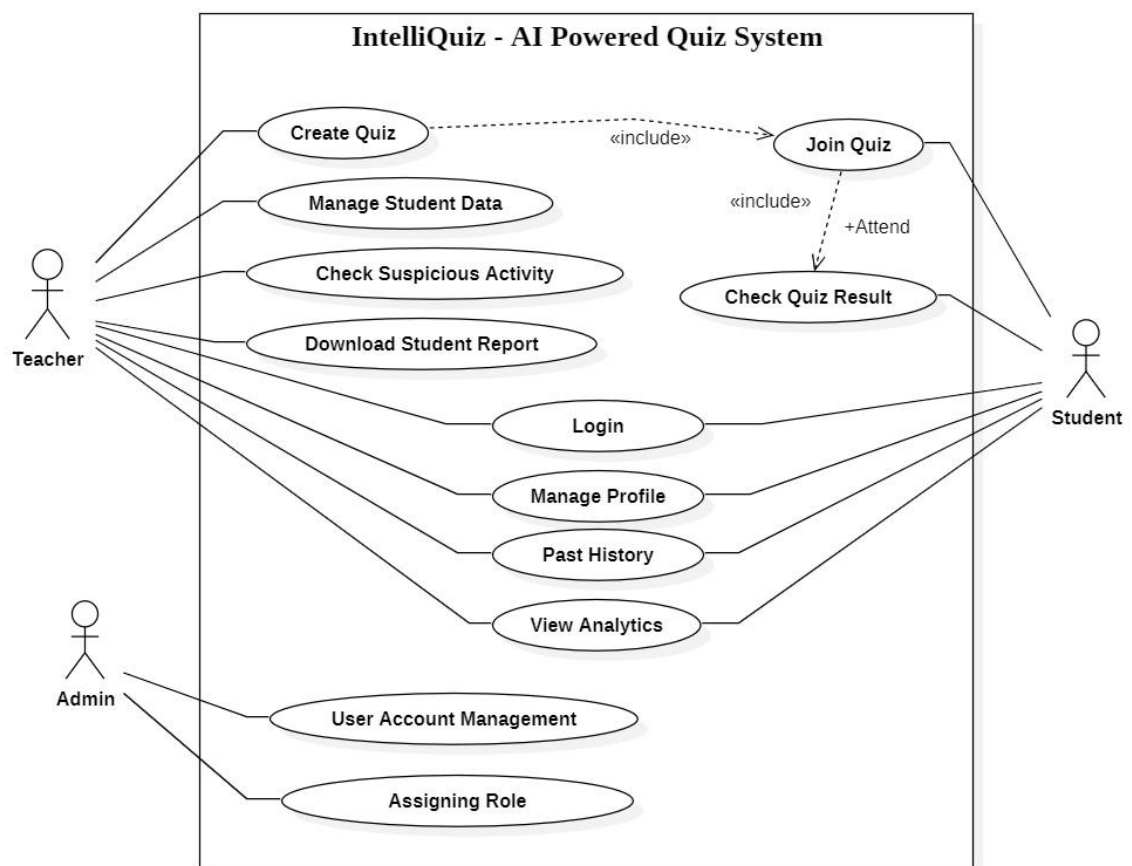


Figure 3 Use Case diagram

As shown in Figure – 3, Use Case Diagram for the IntelliQuiz – AI Powered Quiz System, the diagram illustrates how the Administrator, Teacher, and Student interact with the system’s core functionalities. The Admin manages user accounts and assigns roles, ensuring proper access control. Teachers can create quizzes, manage student data, detect suspicious activities, download reports, and check quiz results, along with shared features like Login, Manage Profile, Past History, and View Analytics. Students mainly use Join Quiz, Check Quiz Result, and other shared functionalities to participate in quizzes and track performance. The diagram also highlights «include» relationships, such as between Create Quiz and Join Quiz, and between Join Quiz and Check Quiz Result, indicating functional dependencies within the system.

7.3.2 Class Diagram

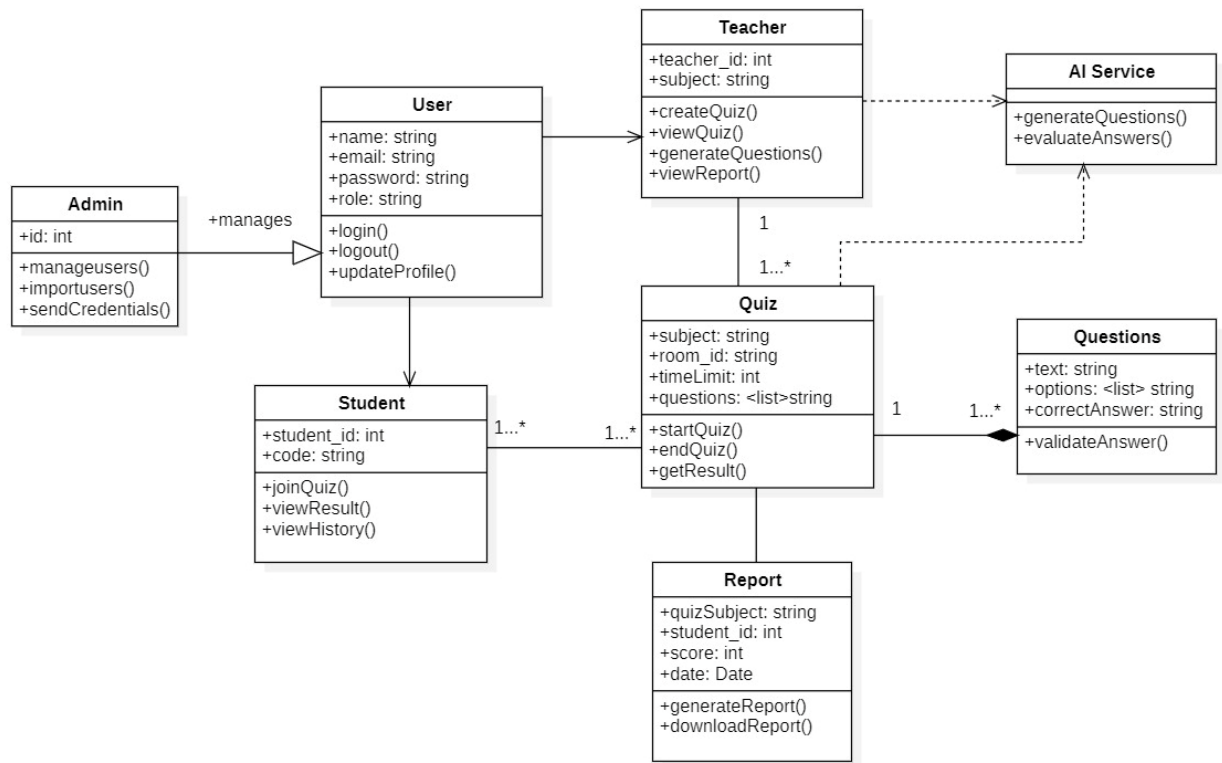


Figure 4 Class diagram

As shown in Figure – 4, the Class Diagram of the IntelliQuiz – AI Powered Quiz System represents the structural design and relationships among key entities such as Admin, User, Teacher, Student, Quiz, Questions, Report, and AI Service. The User class acts as the base class with common attributes like name, email, password, and role, along with methods such as login(), logout(), and updateProfile(). The Admin class extends user functionality through manageUsers(), importUsers(), and sendCredentials(), while the

Teacher class supports createQuiz(), viewQuiz(), generateQuestions(), and viewReport(). The Student class includes operations like joinQuiz(), viewResult(), and viewHistory(), focusing on quiz participation. The Quiz class manages quiz details and operations such as startQuiz(), endQuiz(), and getResult(), whereas the Questions class handles text, options, correctAnswer, and validateAnswer(). The Report class stores quiz performance data and generates downloadable reports, while the AI Service class works with the Quiz and Questions classes for question generation and answer evaluation using Google Gemini AI.

7.3.3 Sequence Diagram

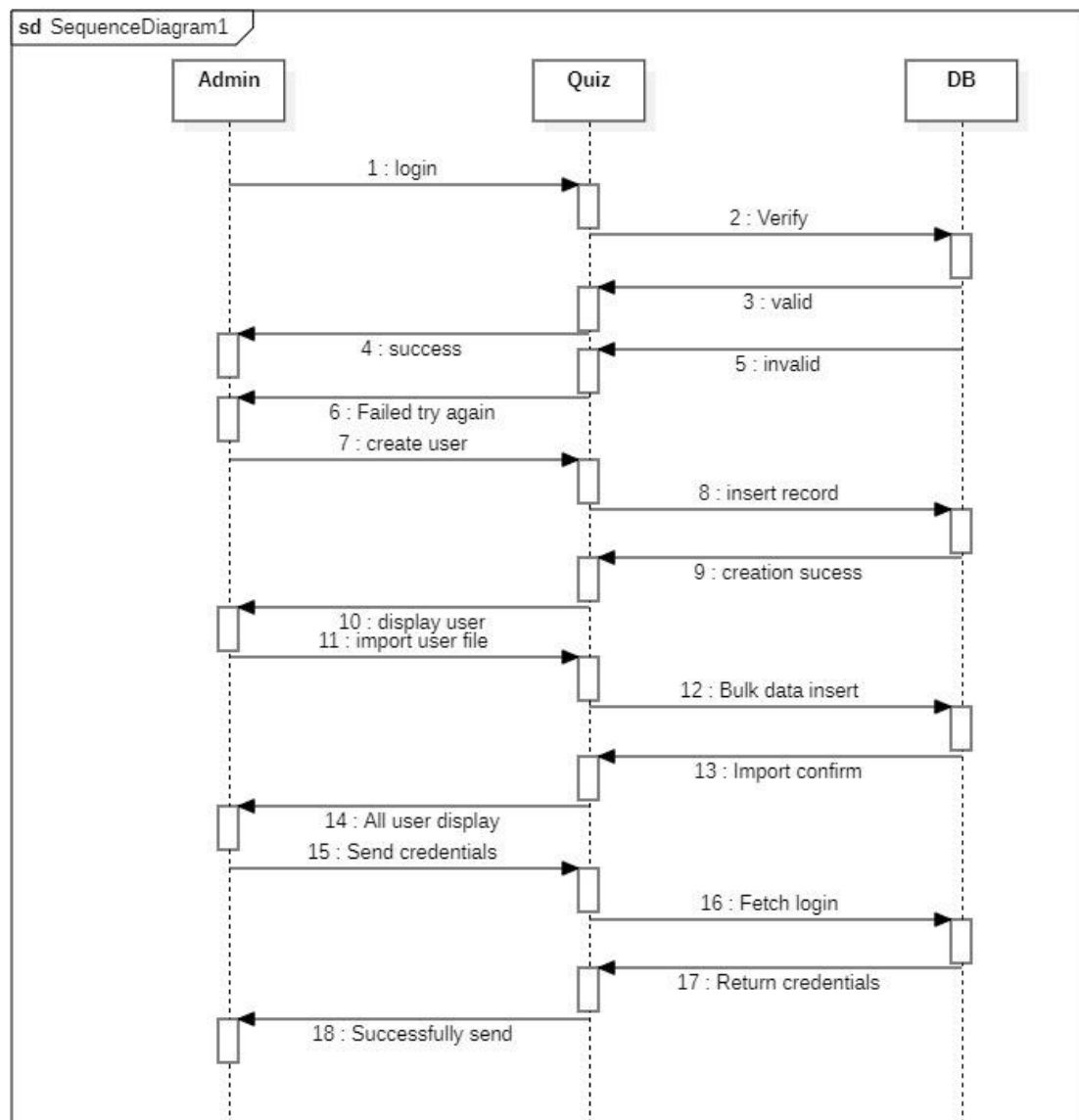


Figure 5 Sequence diagram for Admin

As shown in Figure – 5, the sequence diagram explains the interaction between the Admin, Quiz module, and Database during login, user creation, and data import processes. The Admin

first attempts to log in, which the Quiz module verifies through the database. If valid, the Admin can create users or import bulk user data, which is processed and confirmed by the database. The system then displays all users and allows the Admin to send credentials. Finally, the Quiz module fetches login details from the database and confirms that the credentials are successfully sent.

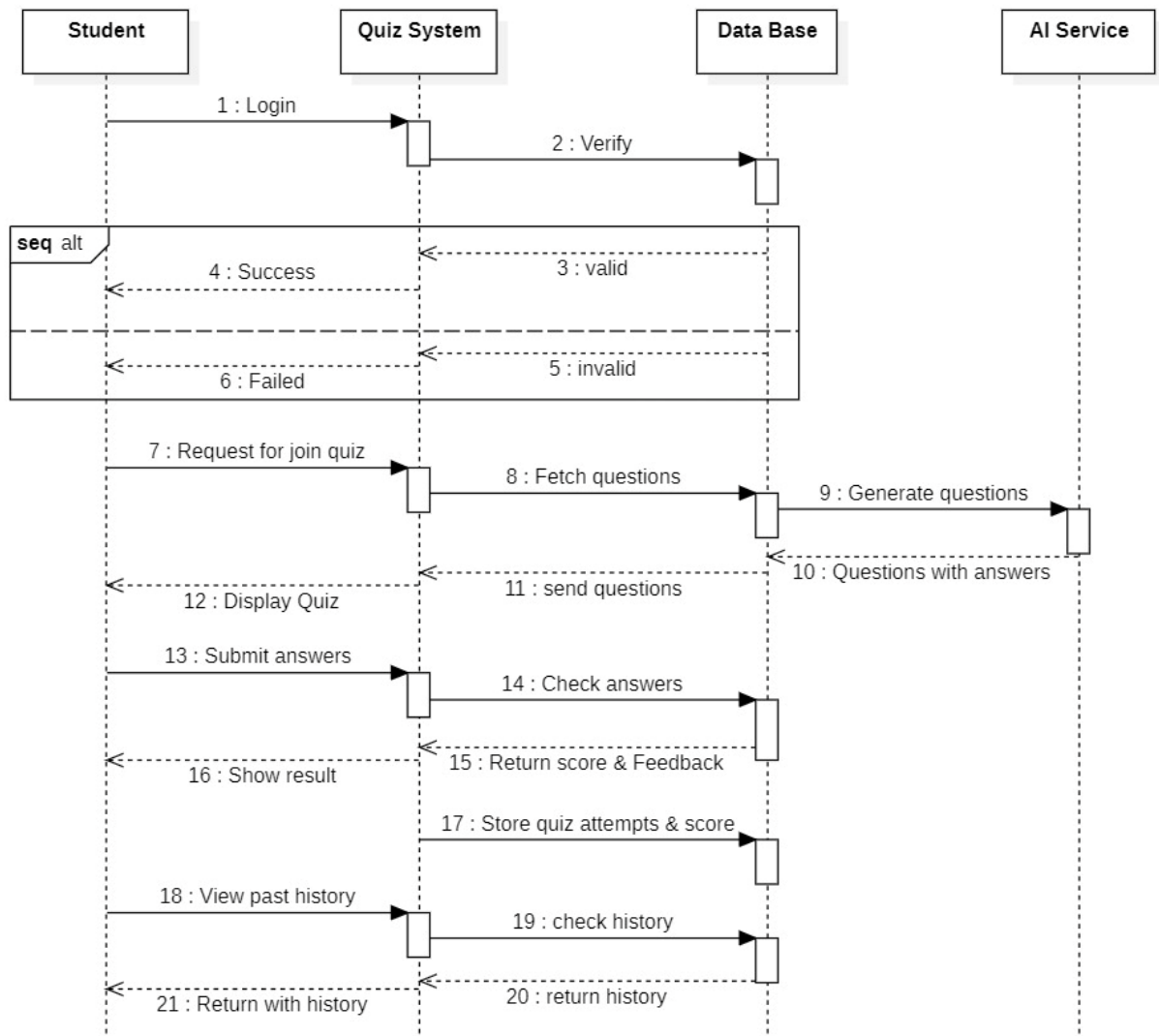


Figure 6 Sequence diagram for Student

As shown in Figure – 6, the sequence diagram illustrates the complete interaction flow of a student using the IntelliQuiz – AI Powered Quiz System. The process begins with the student attempting to log in, followed by the system verifying the credentials through the database and returning either a success or failure response. Upon successful login, the student requests to join a quiz, prompting the system to fetch or generate questions using the AI Service before displaying them. After the student submits answers, the system checks the responses, returns the score and feedback, and stores the quiz attempt in the database.

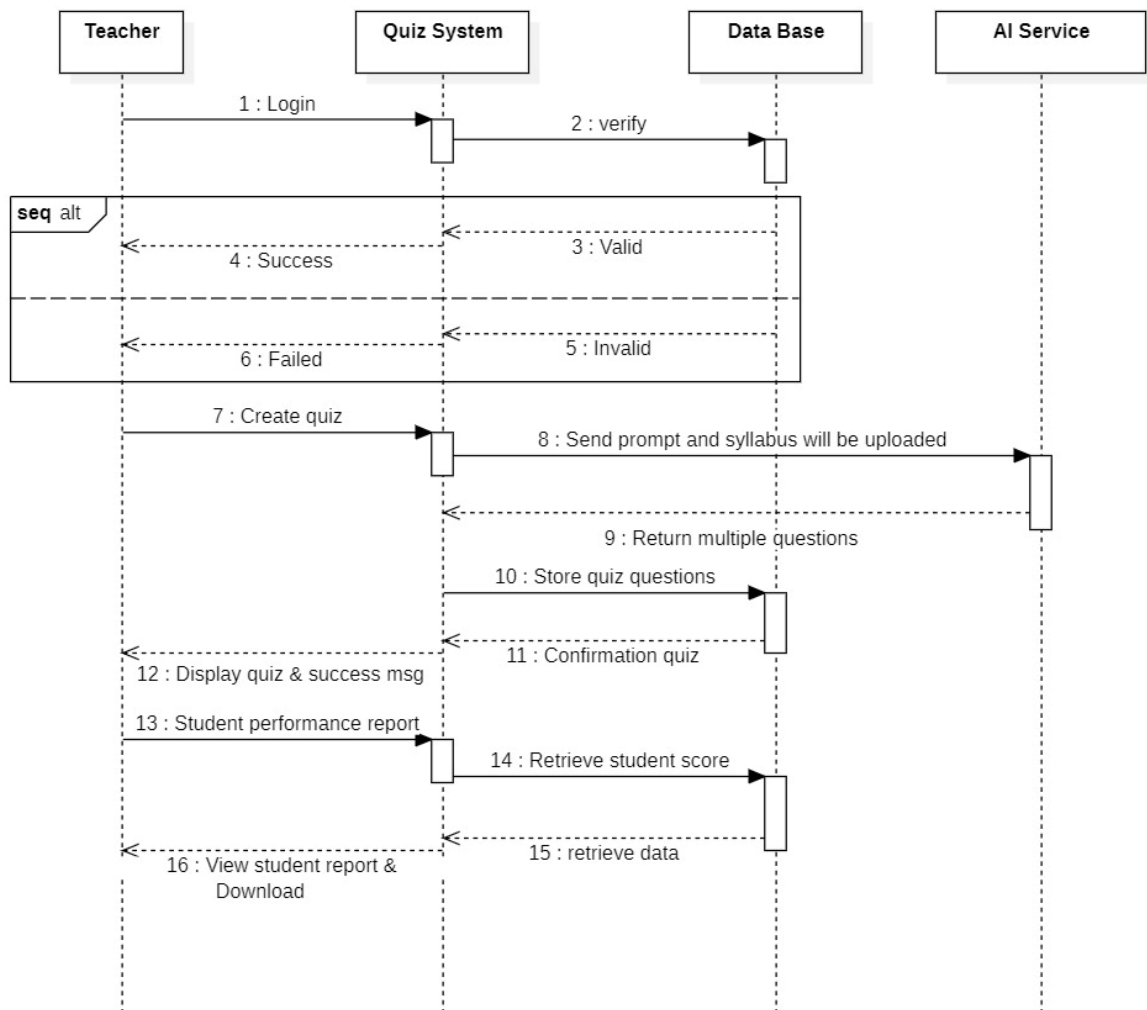


Figure 7 Sequence diagram for Teacher

As shown in Figure – 7, the sequence diagram explains the interaction between the Teacher, Quiz System, Database, and AI Service during quiz creation and report retrieval in the IntelliQuiz system. The Teacher first logs in, and the system verifies credentials through the database, returning either a valid or invalid response. After a successful login, the Teacher creates a quiz, prompting the system to send the prompt and syllabus to the AI Service, which returns multiple AI-generated questions. These questions are then stored in the database, and a confirmation message is shown to the Teacher. The Teacher can later request a student performance report, which triggers the system to retrieve score data from the database and present it for viewing and downloading.

7.3.4 Activity Diagram

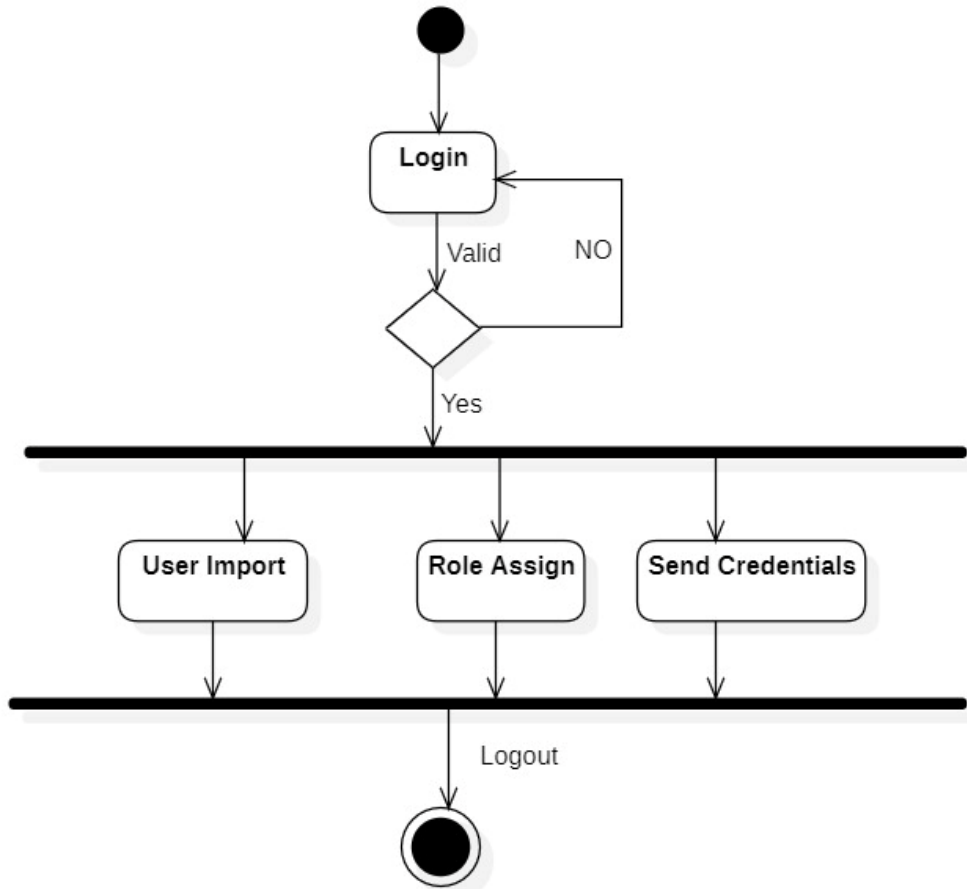


Figure 8 Activity diagram for admin

As shown in Figure – 8 activity diagram driven workflow for user management, beginning with a secure Login decision point that validates credentials; invalid attempts loop back to login while valid authentication proceeds to the main activity flow. Once authenticated, the admin can perform parallel or sequential tasks shown on the action bar, including User Import (bulk CSV upload and database insertion), Role Assign (assigning teacher/student/admin privileges), and Send Credentials (fetching stored login details and dispatching emails). Each action leads to its respective processing and confirmation steps before returning control to the main flow, and the overall process concludes with a Logout action that ends the session. The diagram emphasizes conditional branching at login, clear swimlane separation for administrative tasks, and a clean start-to-end lifecycle that ensures secure, auditable, and efficient user-management operations within the IntelliQuiz platform.

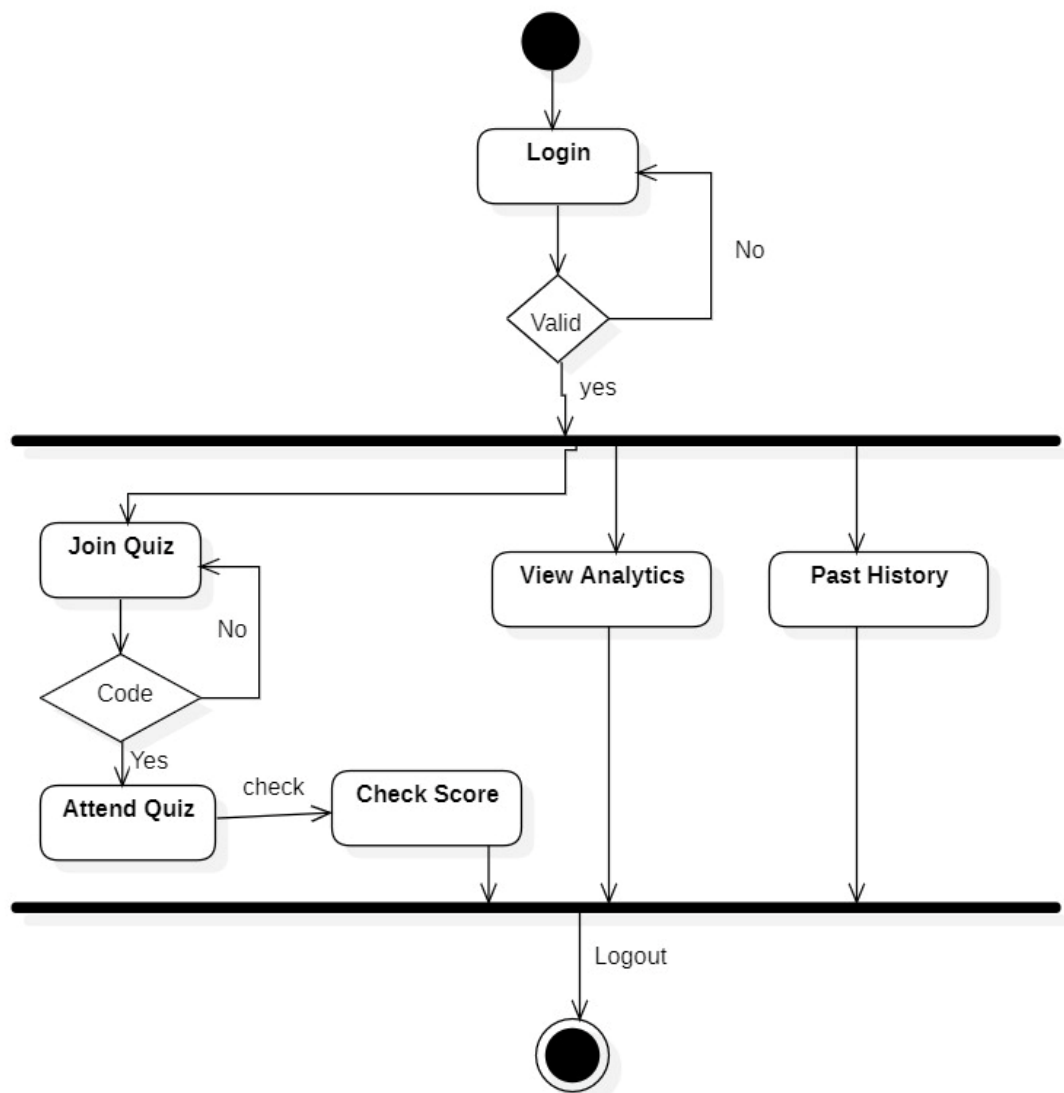


Figure 9 Activity diagram for student

As shown in Figure – 9, the Activity Diagram for the IntelliQuiz – AI Powered Quiz System (Student Module) depicts the complete flow of actions performed by a student on the platform. The process begins with the Login activity, where credentials are validated before granting access to the dashboard. After successful login, the student can choose to Join Quiz, View Analytics, or Check Past History. In the Join Quiz path, the system verifies the quiz code and, if valid, allows the student to attend the quiz, submit responses, and view the AI-evaluated score. Students can also explore analytics to track performance trends or review previous quiz attempts through the Past History option. The activity flow concludes with the Logout action, ensuring secure session termination and smooth system exit. Overall, the diagram highlights a clear and structured sequence integrating authentication, participation, evaluation, and feedback within the student module.

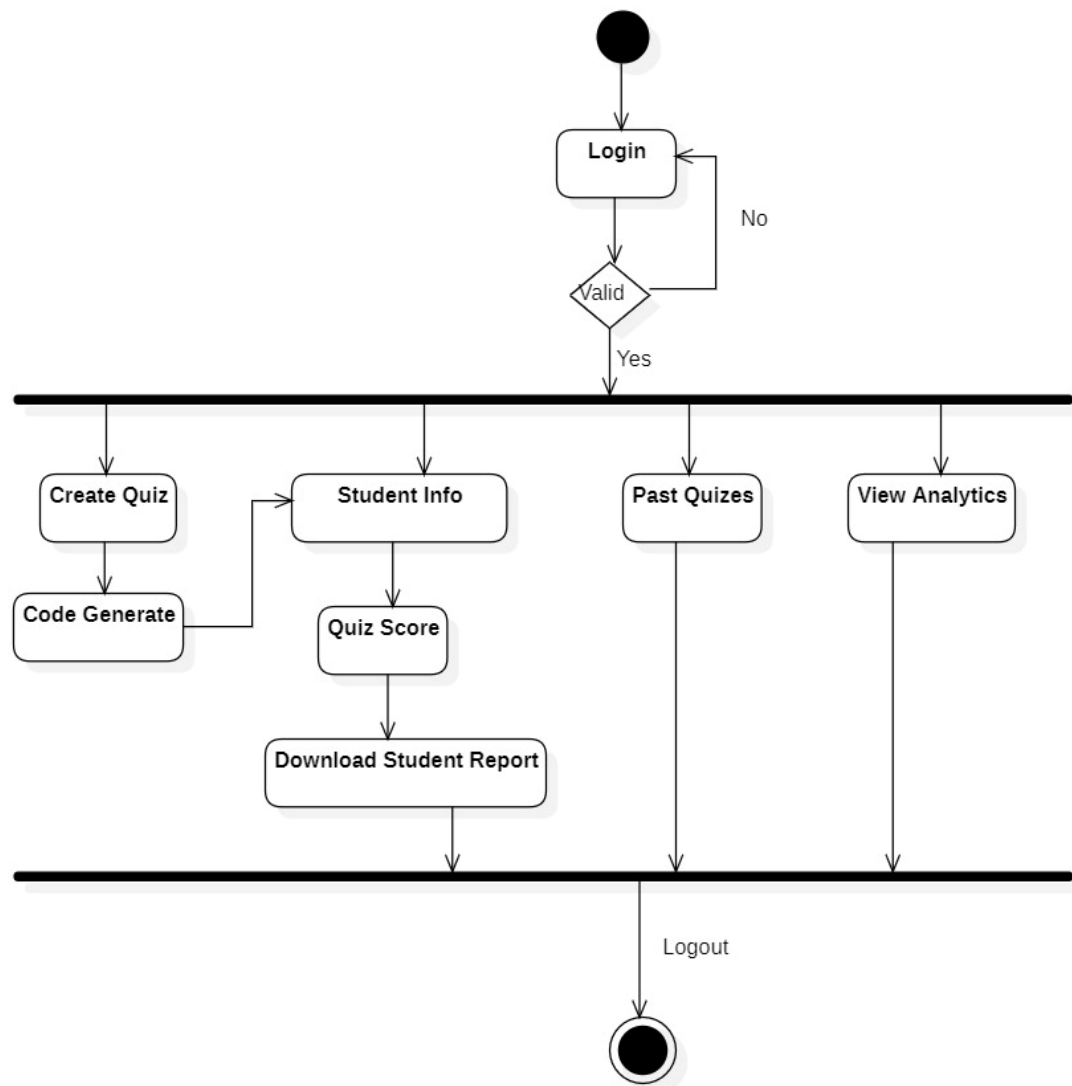


Figure 10 Activity diagram for Teacher

As shown in Figure – 10, the Activity Diagram represents the workflow of a teacher using the IntelliQuiz – AI Powered Quiz System. The process begins with Login, where the system validates credentials and grants access upon successful authentication. Once logged in, the teacher can perform key tasks such as Create Quiz, View Student Info, Check Past Quizzes, and View Analytics. During quiz creation, the system uses Google Gemini AI to generate intelligent, topic-based questions and assign a unique quiz code. Teachers can also view student information, monitor quiz scores, and download reports for analysis. The Past Quizzes and Analytics features help teachers review performance trends and evaluate learning outcomes. The activity flow ends with the Logout action, ensuring secure session closure and proper system exit.

7.3.5 DFD Diagram

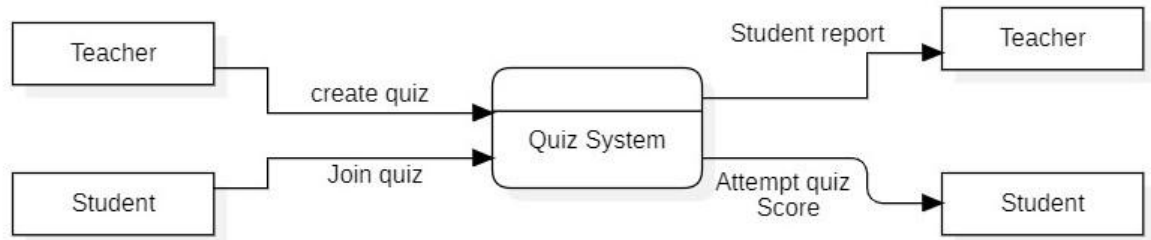


Figure 11 DFD level 0

As shown in Figure – 11, the Level-0 DFD illustrates the overall flow of data among the Teacher, Student, and the Quiz System. The Teacher creates quizzes, which are stored and managed within the system for student access. Students join and attempt these quizzes by submitting their responses, which the system processes to calculate scores. The Quiz System then generates attempt scores for students and detailed performance reports for teachers. These outputs help in evaluation and monitoring of student progress. Overall, the diagram represents a simple yet effective flow that ensures seamless quiz creation, participation, and result generation.

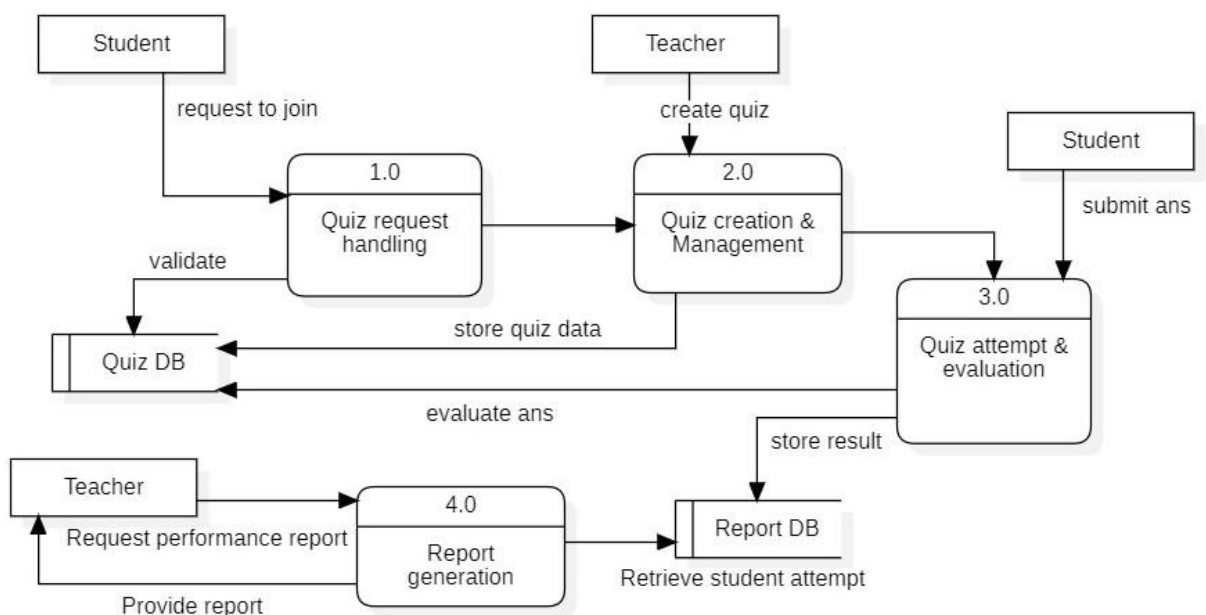


Figure 12 DFD level 1

As shown in Figure – 12, the Level-1 DFD shows how data flows between users, system processes, and databases. The Teacher creates quizzes through the Quiz Creation &

Management module, while Students send join requests that are processed by the Quiz Request Handling component. All quiz details are validated and stored in the Quiz Database. Students then attempt quizzes, and their responses are evaluated automatically by the Quiz Attempt & Evaluation module, with results saved in the Report Database. Finally, the Report Generation module provides teachers with detailed performance reports, ensuring efficient and accurate quiz management.

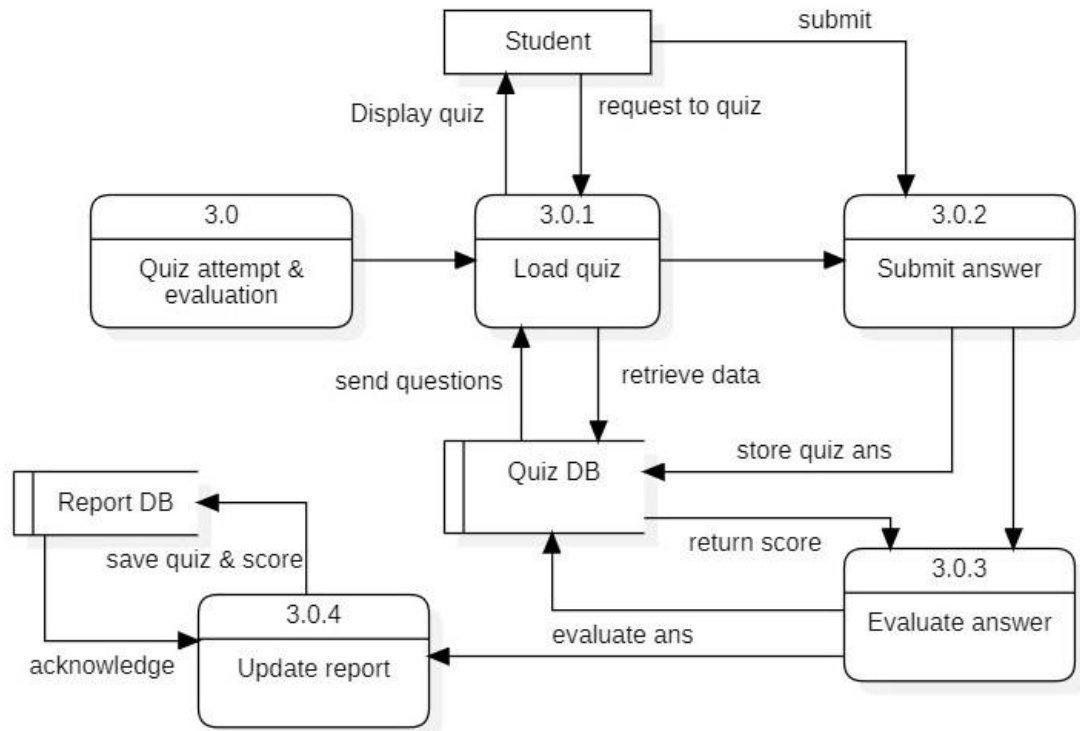


Figure 13 DFD level 2

As shown in Figure – 13, the Level-2 DFD provides a detailed breakdown of the processes within the Quiz Attempt & Evaluation module of the IntelliQuiz system. The flow begins when a student requests to load a quiz (3.0.1), prompting the system to retrieve quiz data from the Quiz Database (Quiz DB) and display it for attempting. After the student submits their answers (3.0.2), the system stores the responses and sends them to the Evaluate Answer process (3.0.3), where the submitted answers are checked against the correct ones, and the score is calculated. The evaluation results are then saved in the Report Database (Report DB) through the Update Report process (3.0.4), ensuring all quiz attempts and performance details are properly recorded. Finally, the system sends an acknowledgment confirming the successful submission, evaluation, and storage of the results. Overall, the diagram clearly outlines the structured and automated workflow of quiz handling, evaluation, and reporting.

7.3.6 ER-Diagram

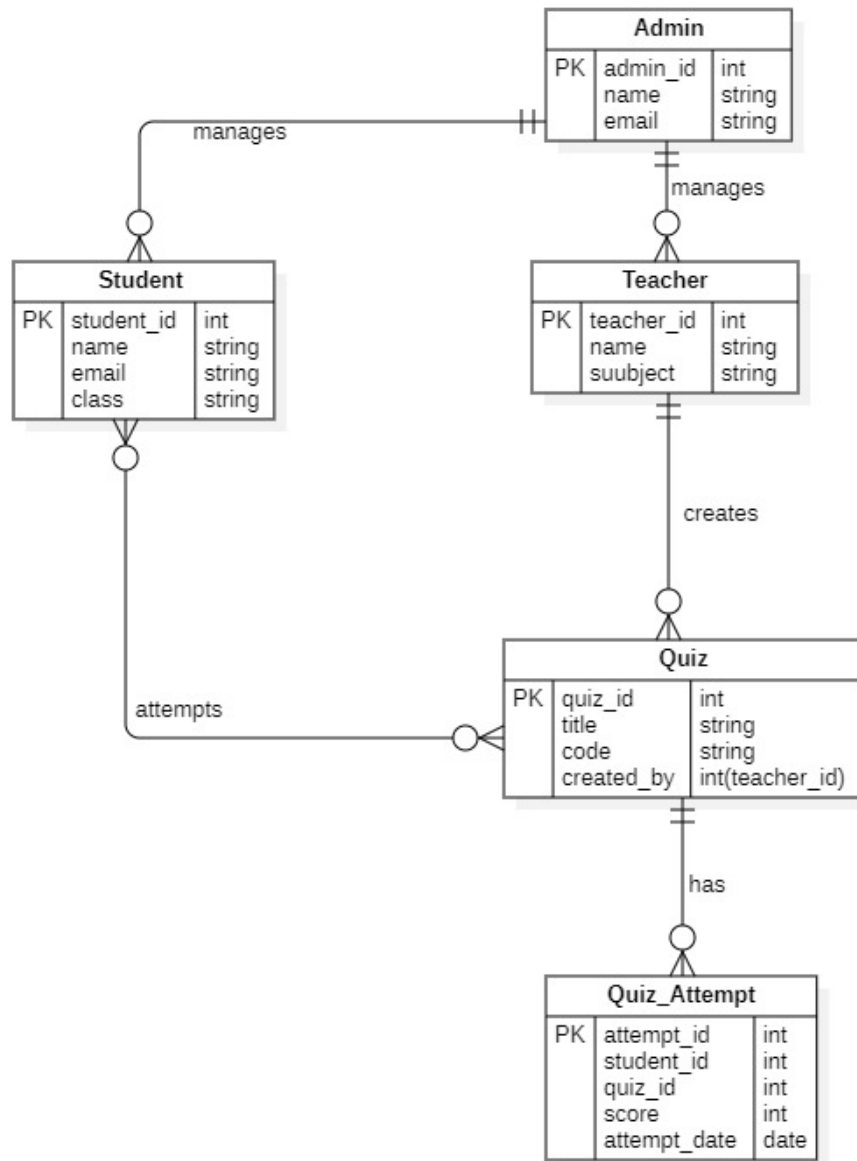


Figure 14 ER Diagram

As shown in Figure – 14, the ER Diagram illustrates the key entities and their relationships within the IntelliQuiz system. The Admin oversees both Teachers and Students, ensuring proper user management and access control. Teachers are responsible for creating quizzes, which are stored in the Quiz entity along with details such as quiz title, room code, and creator information. Students can attempt these quizzes, and every attempt is recorded in the Quiz_Attempt table, storing the student ID, quiz ID, score, and attempt date. The relationships shown between these entities highlight how users, quizzes, and attempts are interlinked to support efficient quiz creation, participation, and result tracking. Overall, the diagram provides a clear structural view of how data is organized and managed across different modules of the system.

7.4 Prototypes

The prototypes for the IntelliQuiz – AI Powered Quiz System illustrate the essential functionalities and interface layouts for different user roles, including Admin, Teacher, and Students. These visual mockups showcase key features such as AI-generated quizzes, secure login, quiz dashboards, performance analytics, and report viewing. They help stakeholders clearly understand the system workflow, user interactions, and overall design structure. By visualizing the interface beforehand, the prototypes ensure better usability.

7.4.1 Prototype for Login Page

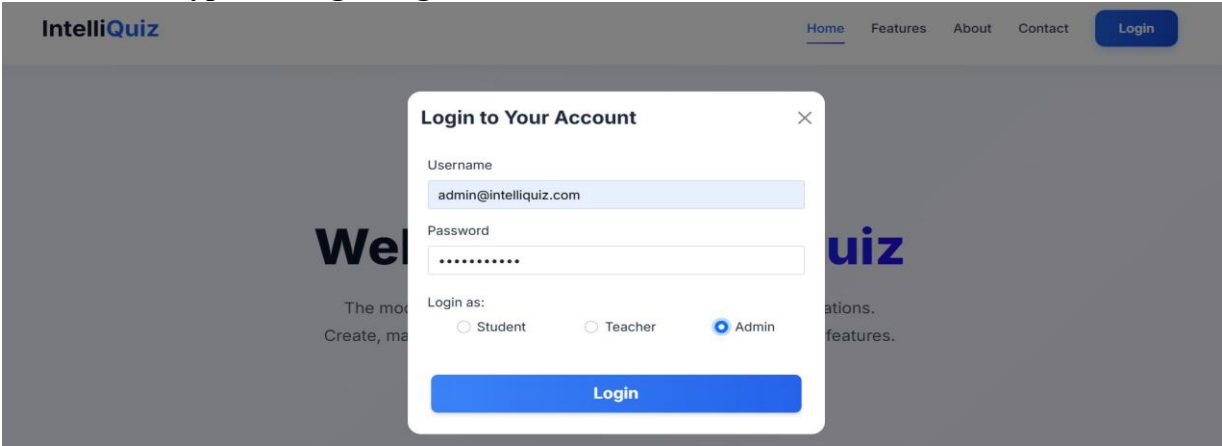


Figure 15 Prototype for Login Page

As shown in Figure – 15 represents the Login Page, which allows all users—Admin, Teacher, and Student—to securely access the IntelliQuiz system using their registered email and password. The page validates user credentials, displays appropriate error messages for invalid login attempts, and ensures secure authentication. Upon successful login, users are automatically redirected to their respective dashboards based on their assigned roles, enabling a personalized and role-specific experience within the system.

7.4.2 Prototype for Admin Dashboard

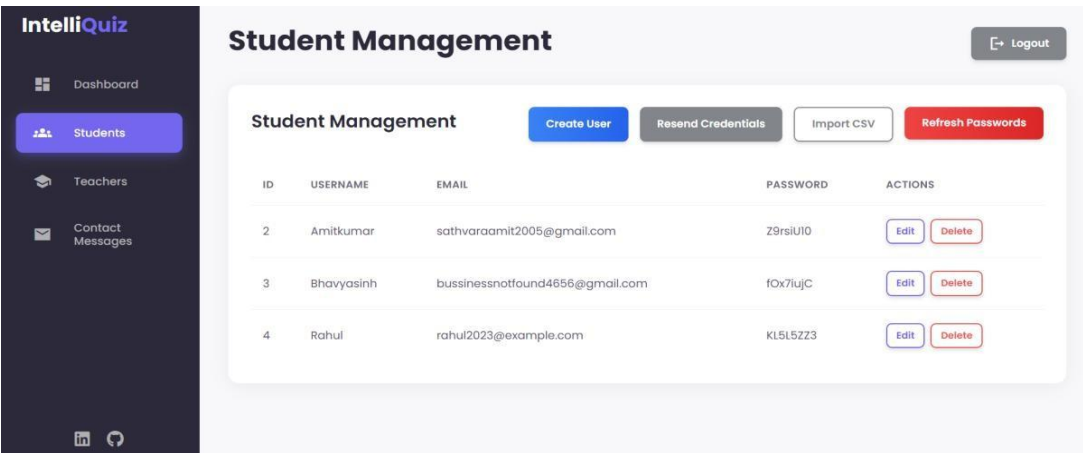


Figure 16 Prototype for Admin Dashboard

As shown in Figure – 16, Admin Dashboard provides a centralized interface for managing all system users. It displays quick navigation cards for adding users, importing data, assigning roles, sending credentials, and viewing the complete user list. It serves as the control center for administrative operations. The Import Users Page allows the admin to upload a CSV file containing bulk student or teacher data. The system processes the file, validates entries, and adds multiple users at once, significantly reducing manual work.

7.4.3 Prototype for Create Quiz Page

1 Quiz Settings — 2 Question Setup

Basic Settings

Quiz Title ⓘ
Python

Number of Questions ⓘ
5

Time Limit (Minutes) ⓘ
2

Quiz Mode ⓘ

⌚

MCQ
Multiple Choice

Security Settings

☒ Enable All Anti-Cheating Features ⓘ

Time Control

- ☒ Strict time limits
- ☒ Auto-submit on timeout
- ☒ Speed monitoring
- ☒ Randomized option order

Browser Security

- ☒ Tab switch detection
- ☒ Full-screen mode
- ☐ Duplicate login detection

Figure 17 Prototype for Create Quiz Page

As shown in Figure – 17, the Create Quiz Page enables teachers to define essential quiz details such as the title, topic, number of questions, and time limit. It features a “Generate Questions with AI” button that automatically fetches topic-based questions using Gemini AI. The page also provides an editable preview of the generated questions, allowing teachers to review and modify content before saving the final quiz. Additionally, teachers can manually adjust or add new questions to ensure accuracy and alignment with learning outcomes.

7.4.4 Prototype for Quiz List Page

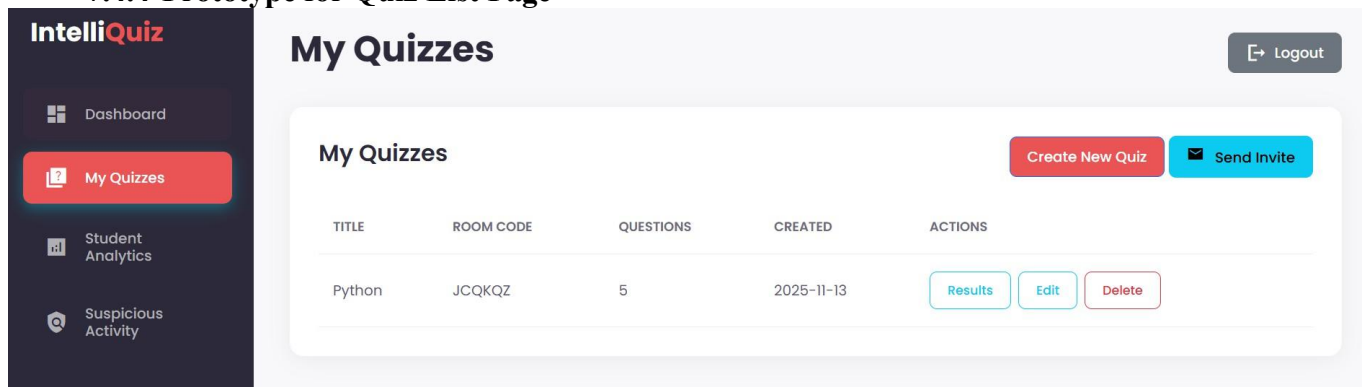


Figure 18 Prototype for Quiz List Page

As shown in Figure – 18, the Quiz List Page displays all quizzes created by the teacher in a structured and organized table format. Each quiz entry includes important details such as the quiz title, room code, creation date, time limit, and total number of questions. Action buttons like View, Edit, Delete, and View Results enable teachers to efficiently manage and monitor their quizzes. This page acts as a central hub for quick access to quiz information, updates, and performance tracking. Overall, it streamlines quiz management and provides clear visibility into student participation across all assessments.

7.4.5 Prototype for Student Participation Monitoring

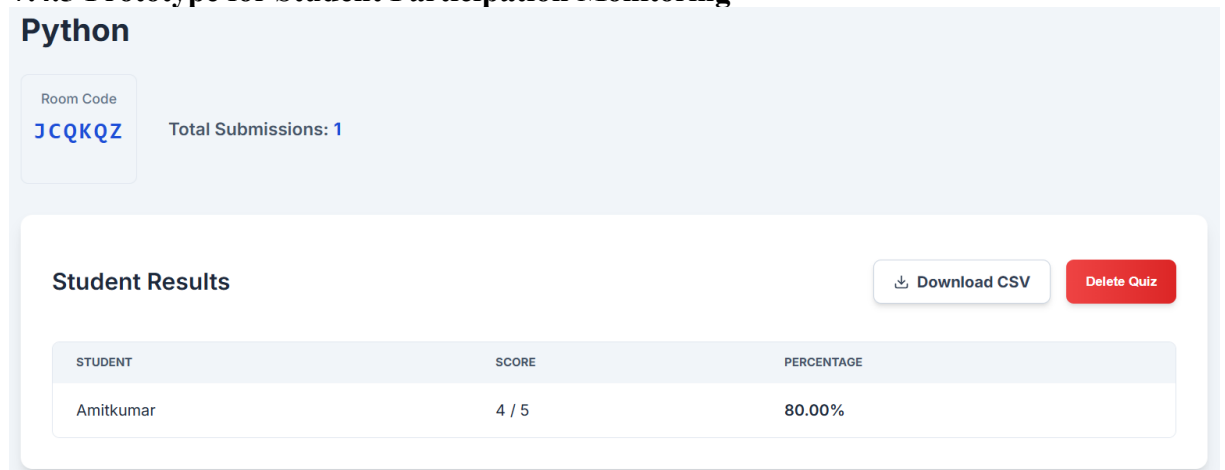


Figure 19 Prototype for Student Participation Monitoring

As shown in Figure – 19, the Results Page displays a student's performance immediately after quiz submission. It shows key details such as the score obtained, total questions attempted, number of correct and incorrect answers, and the time taken to complete the quiz.

7.4.6 Prototype for Activity Log View



Figure 20 Prototype for Activity Log View

As shown in Figure – 20, the Suspicious Activity Page enables teachers to monitor all anti-cheating alerts detected during an ongoing or completed quiz. It displays detailed logs such as tab switching, multiple login attempts, full screen exit, copy-paste actions, and unusual activity timestamps. This page helps teachers identify potential dishonesty, review behavior patterns, and maintain the integrity of online assessments.

7.4.7 Prototype for Join Quiz Page of Student

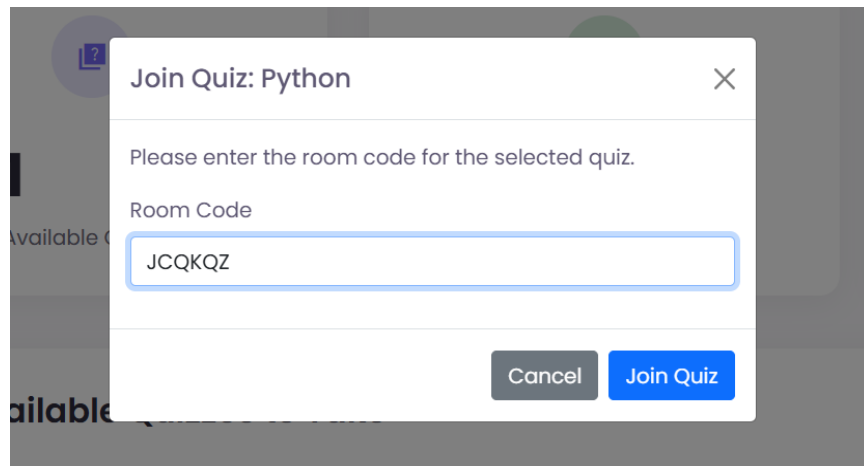


Figure 21 Prototype for Join Quiz Page of Student

As shown in Figure – 21, the Join Quiz Page allows students to enter a room code and connect to an active quiz. The system verifies the entered code and, once validated, displays essential quiz instructions and guidelines. Upon successful verification, the quiz session begins, enabling students to start their attempt seamlessly. This page ensures a smooth and secure entry into the quiz environment.

7.4.8 Prototype for Quiz Attempt

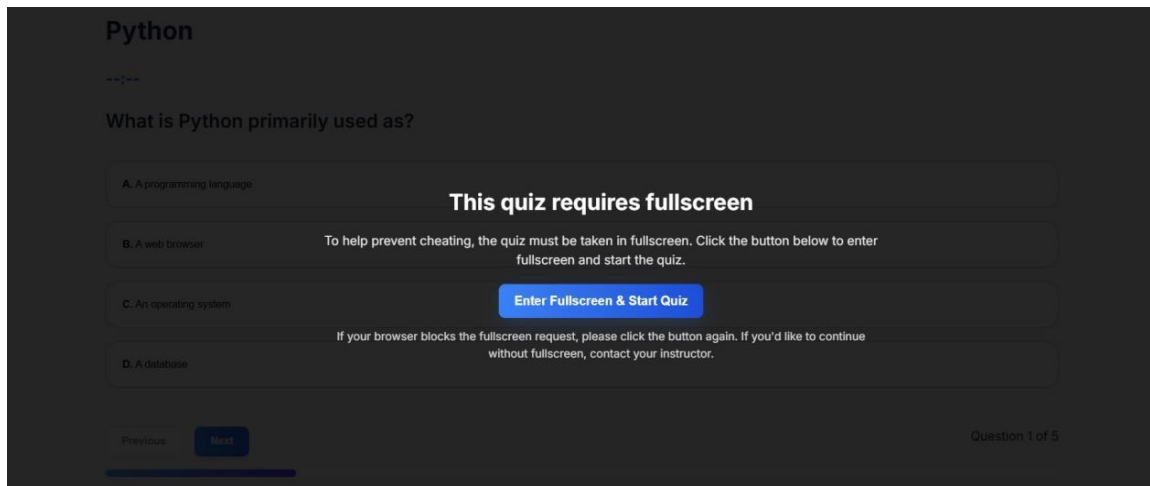


Figure 22 Prototype for Quiz Attempt

As shown in Figure – 22, this prototype shows the full screen requirement popup that appears before a student begins a quiz. The screen informs the student that the quiz can only proceed in full screen and provides a button labeled “Enter Fullscreen & Start Quiz” to begin the attempt. If the browser blocks full screen mode, clear instructions guide the student to retry or contact the instructor. Once full screen is enabled, the quiz interface loads with questions, navigation controls, and a progress indicator for a smooth assessment experience.

7.4.9 Prototype for Quiz Result Page

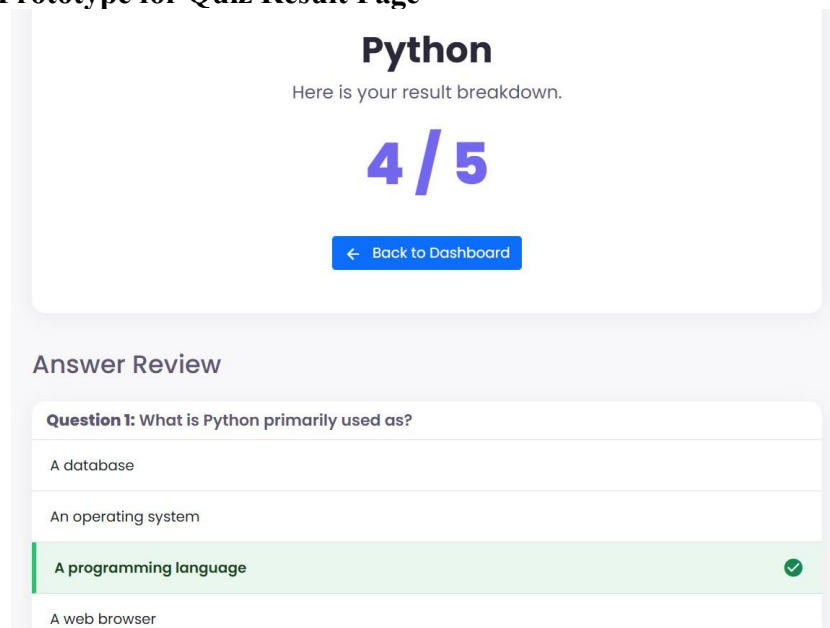


Figure 23 Prototype for Quiz Result Page

As shown in Figure – 23, the Results Page presents a student’s complete performance summary immediately after submitting a quiz. It displays the score obtained, total number of questions, correct and incorrect responses, as well as the time taken to finish the quiz. In addition to individual results, teachers can also access class-wise performance summaries

for broader comparison and evaluation. This interface provides a clear and informative breakdown of student outcomes, supporting effective analysis for both learners and instructors.

7.4.10 Prototype for Past Quiz History Page



Figure 24 Prototype for Past Quiz History Page

As shown in Figure – 24, the Past Quiz History Page allows students to review all previously attempted quizzes in a clear and organized format. It displays key details such as scores, attempt dates, and question-wise performance summaries for each quiz. This feature helps students track their learning progress over time and identify areas that need improvement. Overall, it serves as a valuable reflection tool for continuous academic growth

8. IMPLEMENTATION DETAILS

This chapter provides the technical implementation aspects of the IntelliQuiz – AI Powered Quiz System. It includes the complete flowchart diagram that represents the step-by-step execution flow of core system processes such as login, quiz handling, AI-based evaluation, and result generation. The chapter also documents the source code implementation through a GitHub repository link, enabling developers to access, review, and contribute to the project codebase. Overall, this section bridges the gap between system design and actual development by illustrating workflow logic and providing direct access to implementation resources.

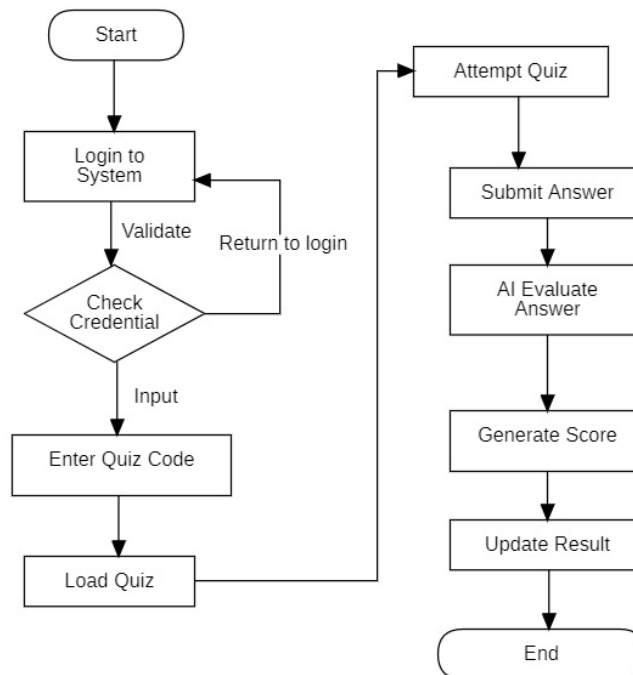


Figure 25 Flowchart for User

As shown in Figure – 25, the flowchart illustrates the complete process followed by a student while attempting a quiz in the IntelliQuiz – AI Powered Quiz System. The sequence begins with the student logging into the system, after which the system validates the entered credentials. If credentials are invalid, the student is prompted to try again until successful authentication. Upon logging in, the student enters the quiz code, and the corresponding quiz is loaded. The student then attempts the quiz and submits their answers, which are evaluated by the integrated AI engine to check correctness and generate the final score. The system updates the result log with the performance data and sends a confirmation notification. The process concludes with the end of the session, completing the quiz attempt workflow.

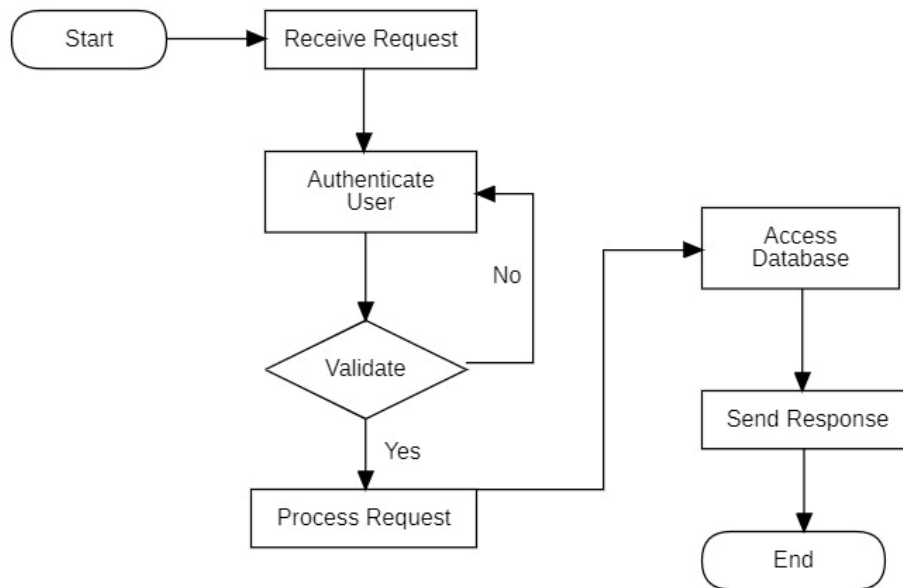


Figure 26 Flowchart Diagram For Server

As shown in Figure – 26, the server-side flowchart illustrates the backend operations involved when a user attempts to mark attendance in the IntelliQuiz – AI Powered Quiz System. After receiving the face image from the client, the server preprocesses it through steps like resizing, normalization, and format conversion. The processed image is then passed to the face recognition or verification model, which compares it with stored facial embeddings to validate the user's identity. If a valid match is found, the server records the attendance, updates the attendance log, and triggers an automated notification confirming successful check-in. If no match is detected, the server returns an error response, prompting the client to retry. This backend workflow ensures secure, accurate, and fully automated attendance handling within the system.



[Sathvara-Amitkumar/IntelliQuiz-AI-Powered-Quiz-System](https://sathvara-amitkumar.github.io/IntelliQuiz-AI-Powered-Quiz-System)

Figure 27 IntelliQuiz – AI Powered Quiz System

As shown in Figure – 27, the QR Code redirects the reader to the official GitHub repositories of the IntelliQuiz – AI Powered Quiz System. These repositories contain all source code, project files, and technical documentation, allowing users and developers to access, review, and download the complete implementation for both the frontend and backend modules.

The IntelliQuiz system is designed to automate and enhance the online quiz-taking process using AI-powered question generation and intelligent evaluation. The Admin can manage user roles and permissions, import or update user details, and monitor system-wide activity. Teachers can create quizzes using AI-generated questions, manage student data, track suspicious activities, and view performance analytics with ease. Students can securely join quizzes, attempt AI-evaluated assessments, and access past quiz history along with detailed performance insights.

The platform ensures accuracy through AI-based answer evaluation and maintains transparency by storing all quiz attempts, analytics, and activity logs. This comprehensive architecture provides a reliable, secure, and user-friendly environment for intelligent assessments.

9. TESTING

Software testing is a crucial phase in the development of the IntelliQuiz - AI Powered Quiz System, aimed at identifying errors, verifying functionality, and ensuring that the software meets all specified requirements. It acts as the final validation step following the stages of requirement gathering, system design, and implementation. For the IntelliQuiz System, testing focuses on verifying the reliability and performance of key functionalities such as user authentication, AI-based quiz generation, quiz participation, result evaluation, and report generation. This section includes comprehensive test cases designed to evaluate the system's accuracy, usability, and robustness under real-world conditions. Each test case specifies the input, expected output, and actual result to confirm the correct behavior of critical modules, including user login, quiz creation, AI-generated question handling, anti-cheating mechanisms, and performance analytics. Both positive and negative test scenarios were executed to assess the system's error handling, data validation, and security responses. The testing outcomes confirmed that the system operates efficiently, securely, and reliably across all user roles—admin, teacher, and student—ensuring that the IntelliQuiz platform fulfills its intended objectives as a dependable and intelligent online assessment system.

9.1 Student Test Cases

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result	Status
ST01	Login with valid credentials	Student has a valid account	1. Enter enrollment number 2. Enter Password 3. Click Login	Student logs in successfully and is redirected to dashboard	Student logged in	Pass

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result	Status
ST02	Login with invalid credentials	Student enters wrong info	1. Enter wrong credentials 2. Click Login	Error message displayed: "Invalid username and Password"	Error displayed	Pass
ST03	Join quiz using valid room code	Student is logged in and quiz is active	Navigate to Join Quiz 1. Enter valid room code	Student successfully joins the quiz interface	Quiz joined successfully	Pass
ST04	Join quiz with invalid room code	Quiz code entered is incorrect	1. Enter invalid code 2. Click Join	System displays "Invalid quiz code" message	Error displayed	Pass
ST05	Attempt quiz and submit answers	Student joined a quiz	1. Answer all questions 2. Click Submit before time expires	Answers submitted successfully and score calculated	Quiz submitted and evaluated	Pass
ST07	View quiz results	Student completed a quiz	1. Navigate to Quiz Results page	Result and score displayed	Results shown	Pass

Table 4 Test cases: Student side

As shown in Table – 4, the test scenarios outline the key functionalities to be validated

for student users in the IntelliQuiz – AI Powered Quiz System. Each test case includes a unique ID, a description of the action being tested, required preconditions, detailed execution steps, the expected system behavior, the actual outcome recorded during testing, and the final pass/fail status. These test cases cover essential student operations such as secure login authentication, joining quizzes, attempting and submitting responses, viewing results, and accessing past performance analytics. Successful execution of these scenarios confirms that the IntelliQuiz system operates as intended, providing students with a secure, reliable, and user-friendly platform for participating in online quizzes and monitoring their progress.

9.2 Teacher Test Cases

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result	Status
TC01	Login with valid credentials	Teacher account exists in the system	1. Enter registered email/username 2. Enter valid password 3. Click Login	Teacher successfully logs in and is redirected to the dashboard	Teacher logged in successfully	Pass
TC02	Login with invalid credentials	None	1. Enter invalid email or password 2. Click Login	System displays an error message indicating invalid credentials	Error displayed	Pass

Test case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result	Status
TC04	Generate AI- based questions	Teacher is logged in and on quiz creation page	1. Enter topic and number of questions 2. Click Generate	System generates accurate topic-based questions using AI	Questions generated correctly	Pass
TC05	Edit or delete a quiz	Teacher has created a quiz	1. Open My Quizzes 2. Select quiz 3. Edit details or click Delete	Quiz is updated or removed successfully	Function performed correctly	Pass
TC06	View student participation	Quiz assigned and students have joined	1. Navigate to Quiz Reports 2. Select quiz 3. View list participants	List of students and their submission statuses displayed	Participant list shown	Pass
TC07	Check quiz results	Quiz completed by students	1. Go to Results section 2. Select quiz 3. View student scores and performance	Results and analytics displayed accurately	Results displayed correctly	Pass

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result	Status
TC08	Monitor cheating activities	Anti-cheating system active	1. Open Activity Log 2. Select quiz 3. Review suspicious activity logs	Any detect cheating attempts are displayed	Logs visible and accurate	Pass
TC09	Download student report	Quiz has been attempted by students	1. Navigate to Reports section 2. Select quiz 3. Click Download Report	Student performance report downloaded in PDF/CSV format	Report downloaded successfully	Pass
TC10	Logout	Teacher is logged in	1. Click Logout on dashboard	System logs out the user and redirects to login page	Logout successfully	Pass

Table 5 Test cases: Teacher side

As shown in Tabel - 5 presents a series of test cases designed to verify the key functionalities accessible to teachers within the IntelliQuiz - AI Powered Quiz System. Each test case is identified by a unique ID and includes a description of the functionality being tested, necessary preconditions, detailed test steps, expected results, actual outcomes, and the final status indicating pass or fail. The tests focus on critical teacher tasks such as secure login authentication, creating and managing quizzes, generating AI-based questions, monitoring student participation and activity logs, and analyzing quiz results and performance reports. These test scenarios ensure that all core features available to teachers operate reliably, supporting efficient quiz creation, student monitoring, and performance evaluation.

9.3 Admin Test Cases

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result	Status
AT01	Admin login with valid credentials	Admin account exists in the system	1. Open login page 2. Enter valid admin credentials 3. Click Login	Admin successfully logs in and is redirected to the admin dashboard	Dashboard loaded Admin logged in successfully	Pass
AT02	Admin login with invalid credentials	None	1. Open login page 2. Enter invalid credentials 3. Click Login	Error message displayed: "Invalid username or password"	Error message shown	Pass
AT03	Add a new user (teacher/student)	Admin is logged in	1. Navigate to User Management 2. Click Add User 3. Fill in user details	New user successfully added to the database	User added successfully	Pass
AT04	Import multiple users via CSV file	Admin has a valid CSV file	1. Go to Import Users 2. Upload CSV file 3. Click Import	Users added in bulk to the database	Users imported successfully	Pass

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result	Status
AT05	Assign roles to users	Users exist in the system	1. Open User Management 2. Select user 3. Choose role 4. Save changes	Role updated successfully and reflected in user list	Role assigned successfully	Pass
AT06	Role assigned successfully	User exists in the system	1. Navigate to User Management 2. Select user	Password reset successfully and notification sent	Password reset and email sent	Pass
AT07	Send login credentials via email	SMTP configured	1. Navigate to Email Notification 2. Select user(s) 3. Click Send Credentials	Login credentials sent successfully to user's email	Emails sent successfully	Pass
AT08	View all registered users	Admin is logged in	1. Go to User List section	All registered users displayed in a table with details	User list displayed	Pass

Table 6 Test cases: Admin side

As shown in Table - 6 lists the important test cases for validating the core administrative functions within the IntelliQuiz - AI Powered Quiz System. Each test case includes a unique ID, a clear description of the functionality being tested, the necessary preconditions, detailed step-by-step instructions, the expected outcomes, actual results, and the final test status. These tests focus on essential admin capabilities such as secure system login, managing user accounts, importing teacher and student data via CSV files, assigning user roles, resetting passwords, and sending login credentials through automated email notifications. Additional test cases also verify the admin's ability to view, update, and delete user information, ensuring complete control over system data. Successfully passing these tests confirms that administrators can efficiently manage users, maintain system security, and oversee platform operations, ensuring the IntelliQuiz System runs smoothly, securely, and effectively across all user levels.

10. USER MANUAL

This chapter provides a step-by-step guide for end users to interact efficiently with the IntelliQuiz - AI Powered Quiz System. It includes clear instructions on how to log in, create or join quizzes, generate AI-based questions, view results, and analyze performance reports. The manual is designed to assist students, teachers, and administrators in navigating the interface and utilizing all key features of the platform. Students learn how to join quizzes using room codes, submit answers, and access their past performance, while teachers are guided through creating quizzes, monitoring participation, and reviewing analytics. Administrators are provided instructions for managing users, assigning roles, and sending login credentials via email. The system's intuitive interface, supported by screenshots and practical examples, helps users understand each function with ease. The primary goal of this manual is to ensure a smooth and user-friendly experience for all users, enabling them to operate the IntelliQuiz System effectively with minimal technical assistance.

10.1 Developer Side Installation Steps

The Developer Side Installation Steps for the IntelliQuiz - AI Powered Quiz System outline the process of setting up the development environment and configuring the system for local or server deployment. To begin, the developer must install Python (version 3.10 or higher) and ensure that pip (Python package manager) is available. Next, clone or download the project source code and navigate to the project directory using the terminal or command prompt. Create a virtual environment using the command `python -m venv venv` and activate it to isolate dependencies. Install all required packages listed in the requirements.txt file using `pip install -r requirements.txt`. Then, configure the .env file with essential environment variables such as the database path, email credentials for SMTP, and API keys for Google Gemini AI integration. Initialize the SQLite database using the provided database.sql script to create all necessary tables. Once setup is complete, run the Flask application with the command `python app.py` or `flask run`, and open the system in a web browser at `http://localhost:5000`. After verifying successful execution, developers can begin modifying features, testing functionalities, or deploying the system on a live server. This process ensures a smooth and standardized setup for development and testing.

10.2 User Side Installation Steps

The User Side Installation Steps for the IntelliQuiz AI-Powered Quiz System are simple and designed to ensure easy access and usability without requiring advanced technical

knowledge. Since the system is web-based, end users—administrators, teachers, and students—do not need to install any additional software. Users only need a device such as a computer, laptop, tablet, or smartphone with a stable internet connection and a modern web browser (e.g., Google Chrome, Mozilla Firefox). The system can be accessed directly through a web link or local network address provided by the institution (e.g., <http://localhost:5000> for local testing or a hosted domain for production use). Upon accessing the platform, users can log in using credentials provided by the administrator. The interface is fully responsive, allowing seamless navigation across devices.

10.3 Login page

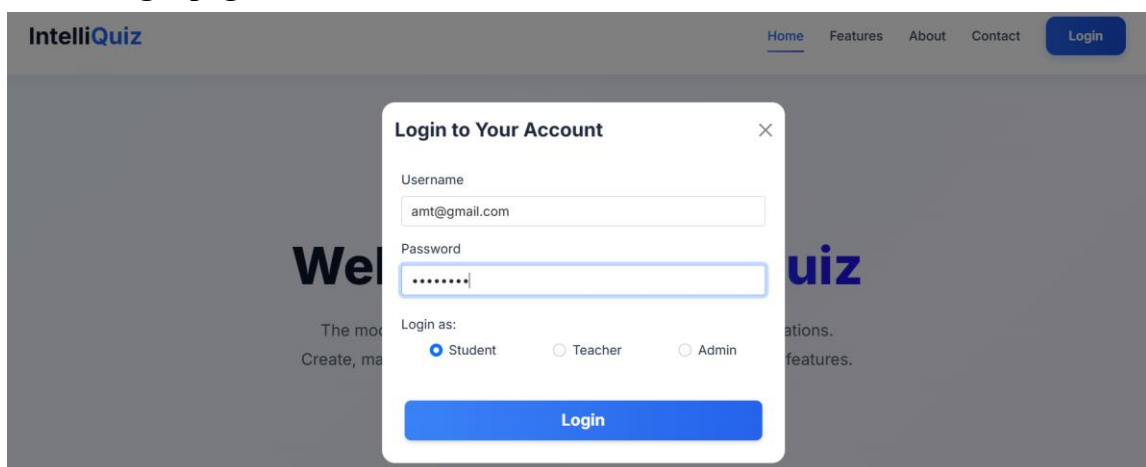


Figure 28 Login page

As shown in Figure – 28, users Student log in using the credentials provided by the administrator. They are required to enter their registered email or enrollment number along with their password. The system validates these inputs, and if the credentials are correct, the user is successfully authenticated.

10.4 Student Dashboard

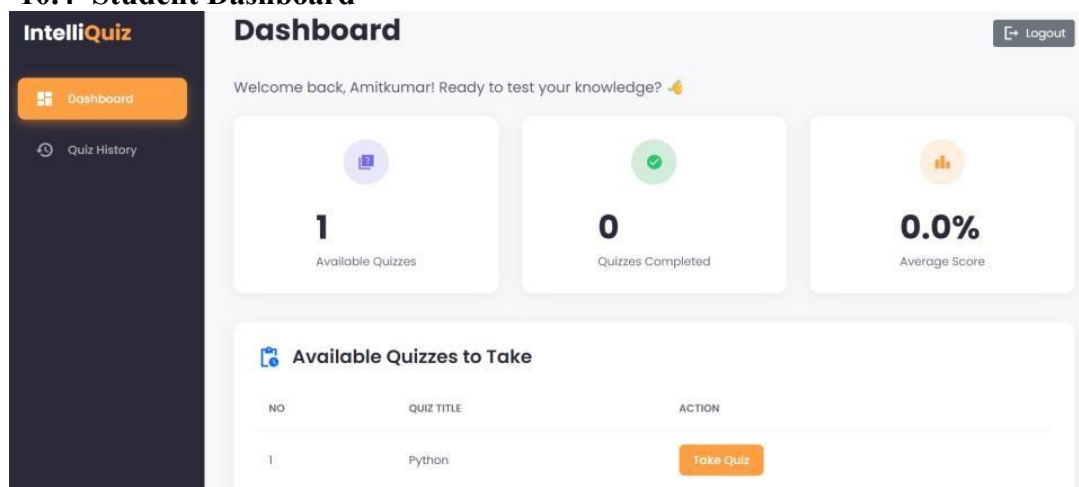


Figure 29 Student Dashboard

As shown in Figure – 29, the Student Dashboard is the central interface where students access all quiz-related features of the IntelliQuiz System. From this dashboard, students can join quizzes using a room code, view instructions, and begin their assessments. They can also check their quiz results, review past quiz history, and explore performance analytics to monitor their progress over time. Designed with simplicity and ease of use in mind, the Student Dashboard allows students to navigate the system effortlessly.

10.5 Student Join Quiz Page

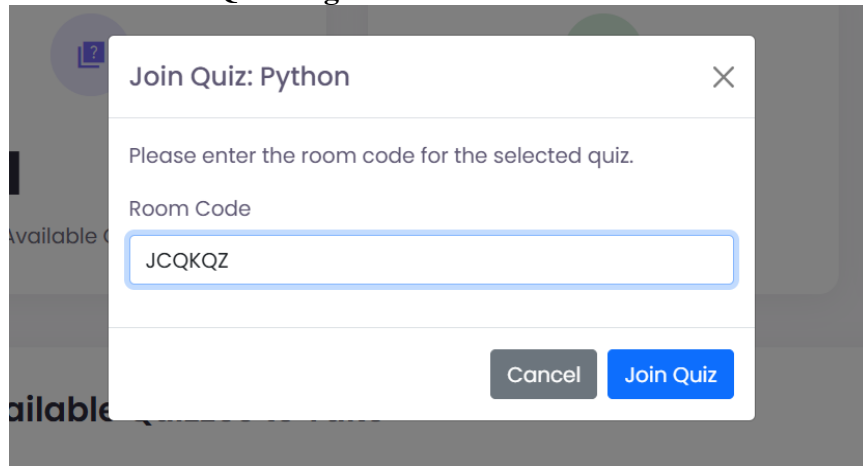


Figure 30 Student Join Quiz Page

As shown in Figure – 30, the Join Quiz Page allows students to enter a room code and connect to an active quiz. The system verifies the entered code and, once validated, displays essential quiz instructions and guidelines. Upon successful verification, the quiz session begins, enabling students to start their attempt seamlessly.

10.6 Student Quiz Attempt Page

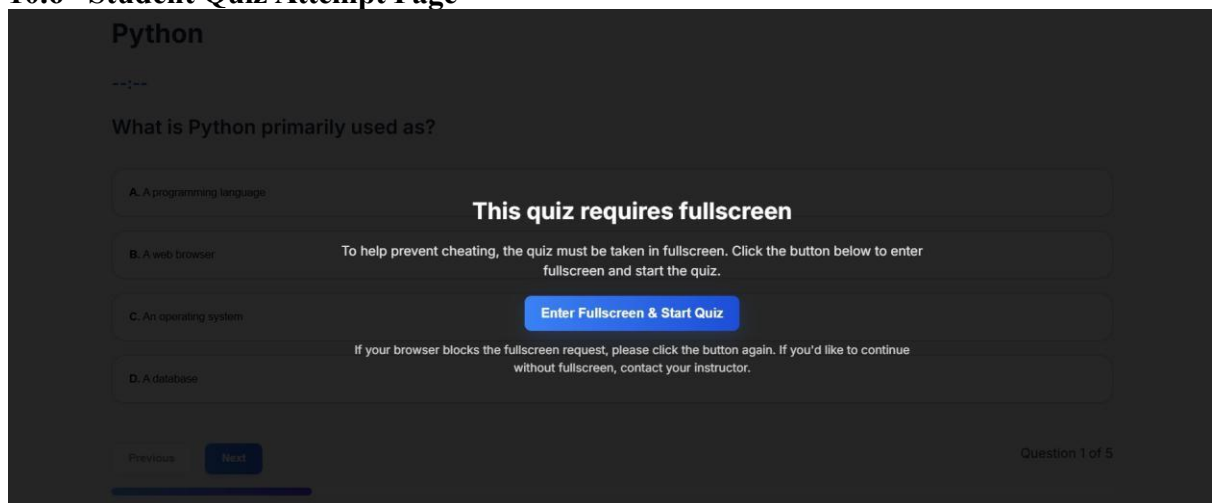


Figure 31 Student Quiz Attempt Page

As shown in Figure – 31, this prototype shows the full screen requirement popup that

appears before a student begins a quiz. To ensure exam integrity, the system enforces full screen mode and restricts any form of navigation outside the quiz window. The screen informs the student that the quiz can only proceed in full screen and provides a button labeled “Enter Fullscreen & Start Quiz” to begin the attempt. If the browser blocks full screen mode, clear instructions guide the student to retry or contact the instructor.

10.7 Quiz Result Page

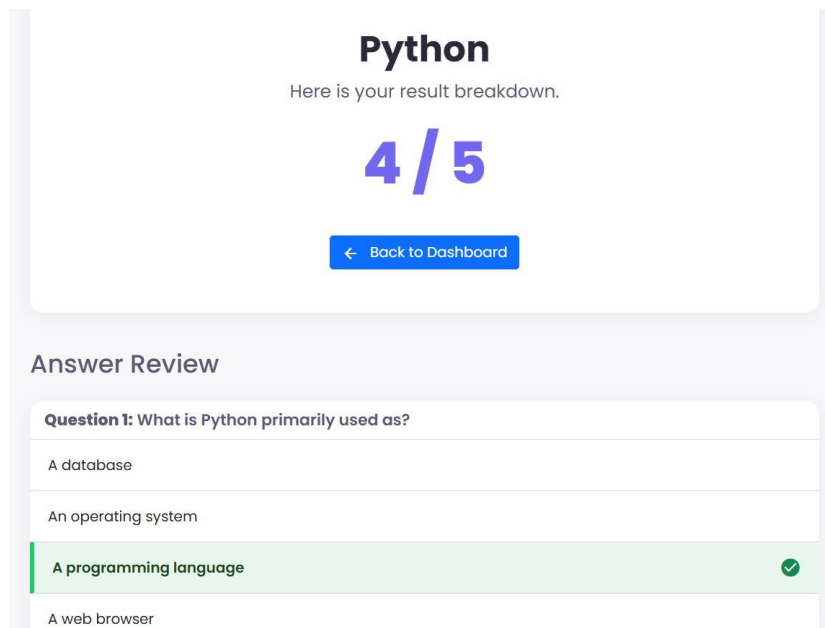


Figure 32 Quiz Result Page

As shown in Figure – 32, the Results Page presents a student’s complete performance summary immediately after submitting a quiz. It displays the score obtained, total number of questions, correct and incorrect responses, as well as the time taken to finish the quiz. In addition to individual results, teachers can also access class-wise performance summaries for broader comparison and evaluation. This interface provides a clear and informative breakdown of student outcomes, supporting effective analysis for both learners and instructors.

10.8 Past Quiz History Page



Figure 33 Past Quiz History Page

As shown in Figure – 33, the Past Quiz History Page allows students to review all previously attempted quizzes in a clear and organized format. It displays key details such as scores, attempt dates, and question-wise performance summaries for each quiz. This feature helps students track their learning progress over time and identify areas that need improvement. Overall, it serves as a valuable reflection tool for continuous academic growth.

11. CONCLUSION AND FUTURE WORK

The IntelliQuiz - AI Powered Quiz System provides an intelligent, secure, and efficient solution for automating online assessments through the integration of artificial intelligence and web technologies. By using Google Gemini AI, the system simplifies quiz creation, ensures question diversity, and reduces manual workload for teachers while maintaining fairness through built-in anti-cheating mechanisms such as randomization, time limits, and activity tracking. The platform's role-based structure allows administrators, teachers, and students to interact seamlessly, supported by a user-friendly interface and a lightweight Flask-SQLite architecture. Overall, the system enhances digital learning by streamlining quiz management, evaluation, and result analysis. For future work, the system can be expanded with machine learning-based performance analytics, LMS integration, cloud database deployment, and mobile application support to improve scalability, accessibility, and adaptability. These enhancements would make IntelliQuiz a more robust, intelligent, and comprehensive solution for modern education and e-assessment systems..

11.1 Future Work

In the future, the IntelliQuiz - AI Powered Quiz System can be enhanced with several advanced features to improve its intelligence, scalability, and user experience. One of the key upgrades will be the integration of machine learning analytics to automatically evaluate student performance trends, predict learning gaps, and generate personalized quiz recommendations for adaptive learning. To strengthen data security, the system can incorporate multi-factor authentication (MFA)—requiring users to verify their identity via OTP, email verification, or biometric authentication—ensuring secure access and preventing unauthorized entry. Another significant improvement will involve cloud-based deployment using platforms like AWS or Google Cloud, enabling better scalability, faster performance, and support for real-time multi-user access across large institutions. Additionally, a dedicated cross-platform mobile application will be developed to allow teachers and students to create, attempt, and review quizzes directly from their smartphones, complete with push notifications for quiz schedules, result announcements, and performance updates. The system can also be integrated with Learning Management Systems (LMS) and academic databases to automate quiz assignments and grade synchronization. Future versions may also include voice-based quiz interaction, AI-powered plagiarism detection, and real-time proctoring through webcam monitoring to ensure academic integrity. Furthermore, a data visualization

dashboard will be added for educators to analyze class performance and generate detailed reports effortlessly. These advancements will make IntelliQuiz a more intelligent, secure, and adaptive e-learning platform, capable of transforming digital education and assessment management.

ANNEXURE

This section provides clear definitions and explanations of the key technical terms, components, and technologies used in the development of the IntelliQuiz – AI Powered Quiz System, helping readers understand the underlying concepts and implementation framework.

Flask: A lightweight Python web framework used to build the backend of the system, offering simplicity, flexibility, and scalability.

Google Gemini AI: A powerful artificial intelligence model integrated into the system for generating dynamic, topic-based quiz questions automatically.

SQLite: A lightweight, serverless database system used for storing user information, quizzes, questions, and results locally.

SMTP (Simple Mail Transfer Protocol): A communication protocol used to send automated emails such as login credentials and quiz notifications.

HTML/CSS/JavaScript: Core web technologies used for developing the frontend interface, ensuring a responsive and user-friendly experience.

RESTful API: A design principle used for structuring communication between the frontend and backend, ensuring smooth data transfer.

RBAC (Role-Based Access Control): A security model that restricts access and permissions based on user roles—Administrator, Teacher, or Student.

AI Question Generation: A feature that uses Google Gemini AI to automatically create quiz questions based on specific topics or difficulty levels.

Version Control (Git/GitHub): A tool for managing source code versions, enabling collaboration among developers.

Deployment Platforms (Render/Vercel): Cloud-based hosting services used for deploying backend (Flask) and frontend applications.

Encryption: A data security technique used to protect user credentials and sensitive information stored in the database.

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TOOLS AND TECHNOLOGIES

Frontend Technologies

1. **HTML, CSS, JavaScript:** For developing the basic structure, design, and interactivity of the web interface.

Backend Technologies

1. **Flask (Python Framework)** – A lightweight yet powerful backend framework used to build REST APIs for authentication, quiz creation, AI question generation, anti-cheating logs, and result management.
2. **Python Libraries (Flask-CORS, SQLite, Requests)** – Used for handling API communication, database integration, and external AI service calls.

AI and Question Generation

1. **Google Gemini API** – Used to automatically generate quiz questions based on the topic provided by teachers. Handles text-based AI output and assists in generating MCQs, explanations, and variations.

Database

1. **SQLite** – Lightweight relational database used to store user accounts, quizzes, questions, results, suspicious activity logs, and student performance data.

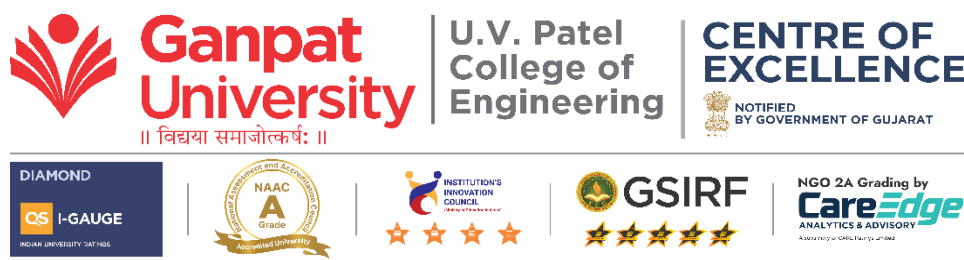
Development Tools

1. **Visual Studio Code:** A source-code editor used for frontend and backend development.
2. **Git & GitHub:** For version control and collaboration.

Deployment Platforms

1. **Vercel:** Used for hosting the React frontend for quick and reliable access

ABOUT COLLEGE



Ganpat University-U. V. Patel College of Engineering (GUNI-UVPCE) is situated in Ganpat Vidyanagar campus. It was established in September 1997 with the aim of providing educational opportunities to students from It is one of the constituent colleges of Ganpat University various strata of society. It was armed with the vision of educating and training young talented students of Gujarat in the field of Engineering and Technology so that they could meet the demands of Industries in Gujarat and across the globe.

The College is named after Shri Ugarchandbhai Varanasibhai Patel, a leading industrialist of Gujarat, for his generous support. It is a self-financed institute approved by All India Council for Technical Education (AICTE), New Delhi and the Commissionerate of Technical Education, Government of Gujarat.

The College is spread over 25 acres of land and is a part of Ganpat Vidyanagar Campus. It has six ultra-modern buildings of architectural splendor, class rooms, tutorial rooms, seminar halls, offices, drawing hall, workshop, library, well equipped departmental laboratories, and several computer laboratories with internet connectivity through 1 Gbps Fiber link, satellite link education center with two-way audio and one-way video link. The superior infrastructure of the Institute is conducive for learning, research, and training. The Institute offers various undergraduate programs, postgraduate programs, and Ph.D. programs.